

Forward Manipulator Kinematics

Chapter 3

MAE C163A/C263A

Forward Manipulator Kinematics

Introduction

Kinematics

The science of motion that treats motion *without regard to the forces* which cause it (e.g. position, velocity, acceleration, higher derivatives of the position).

Kinematics of Manipulators

The study of all the geometrical and time based properties of the motion.

Forward Manipulator Kinematics

Central Topic

Problem

Given:

The manipulator's geometrical parameters
(dimensions, configuration, joint angles, etc.)

Specify:

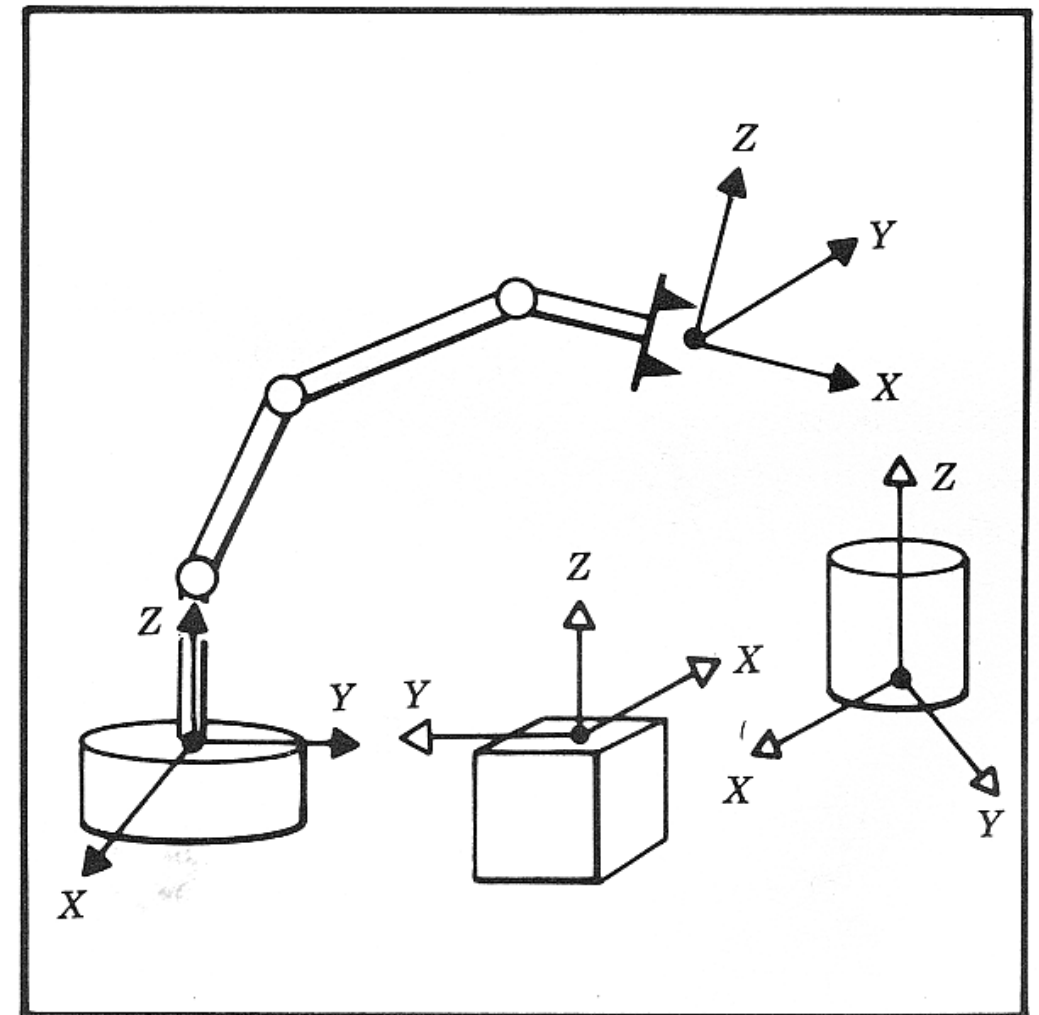
The position and orientation of the manipulator

Solution

Coordinate systems or “Frames” are attached to the manipulator and objects in the environment following the Denavit-Hartenberg notation (“DH Parameters”)

Forward Kinematics is also called “Direct Kinematics”

The “DH” in “DH Parameters” is from “Dennis Hong” Parameters (kidding...)



Forward Manipulator Kinematics

Kinematic Chain

Kinematic Chain

A kinematic Chain consists of nearly rigid **links** (members) which are connected by **joints** (kinematic pair) that allow relative motion of neighboring links.

Closed Loop Chain

Every link in the kinematic chain is connected to any other link by at least two distinct paths

Open Loop Chain

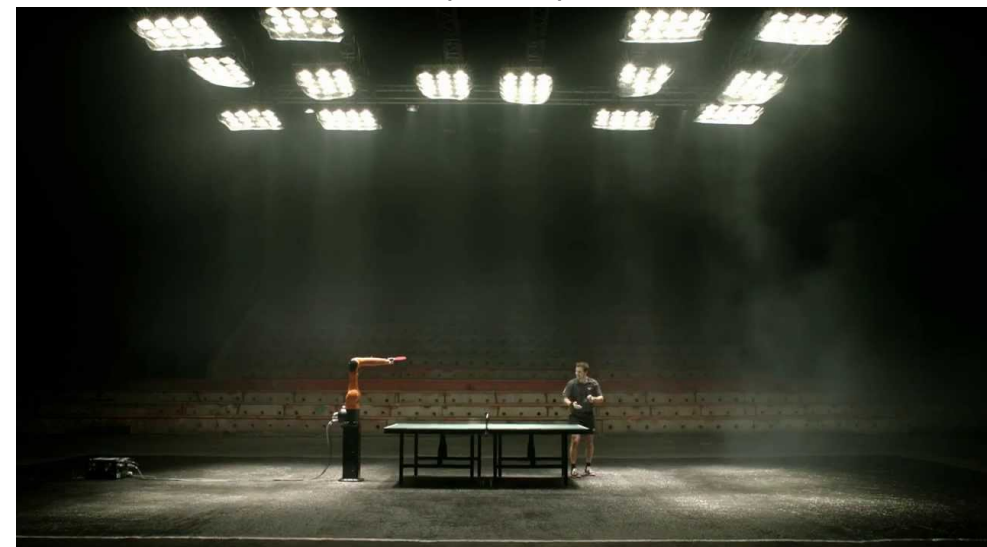
Every link in the kinematic chain is connected to any other link by one and only one distinct path

(Video)



Parallel (Close Loop) Robot

(Video)



Serial (Open Loop) Robot

Forward Manipulator Kinematics

Degrees of Freedom (DOF)

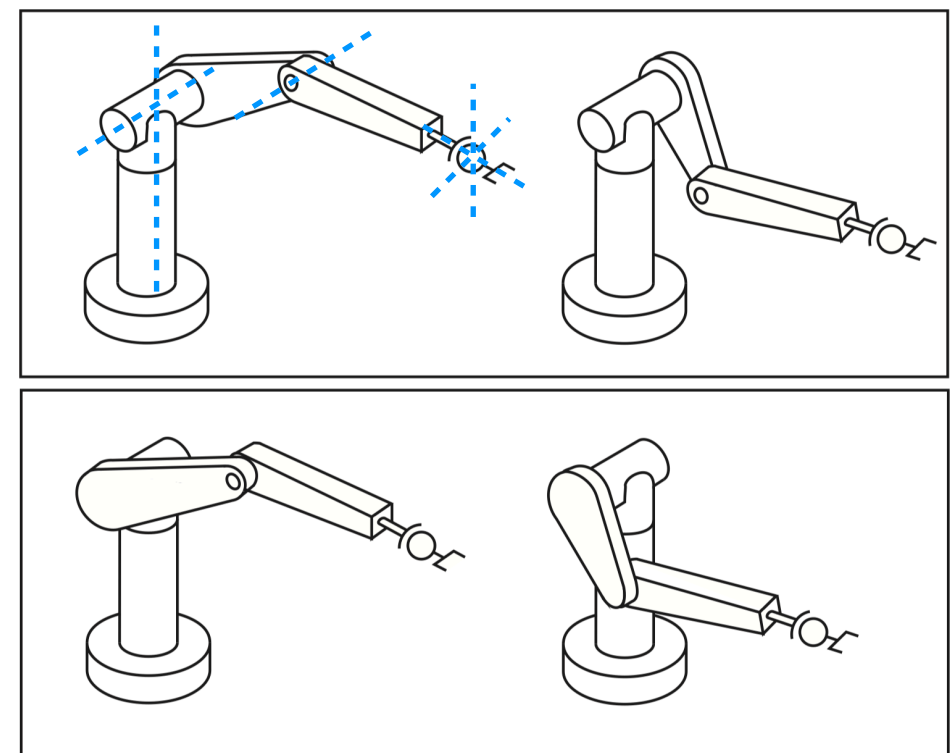
The number of *Degrees of Freedom* that a manipulator possesses is the number independent position variable which would have to be specified in order to locate all parts of the mechanism.

Ideally, a manipulator should poses 6 DOF in order to manipulate an object in a 3D space

General Purpose Robot: # DOF = 6

Redundant Robot: # DOF > 7

Deficient Robot: # DOF < 6



6 DOF PUMA arm

Forward Manipulator Kinematics

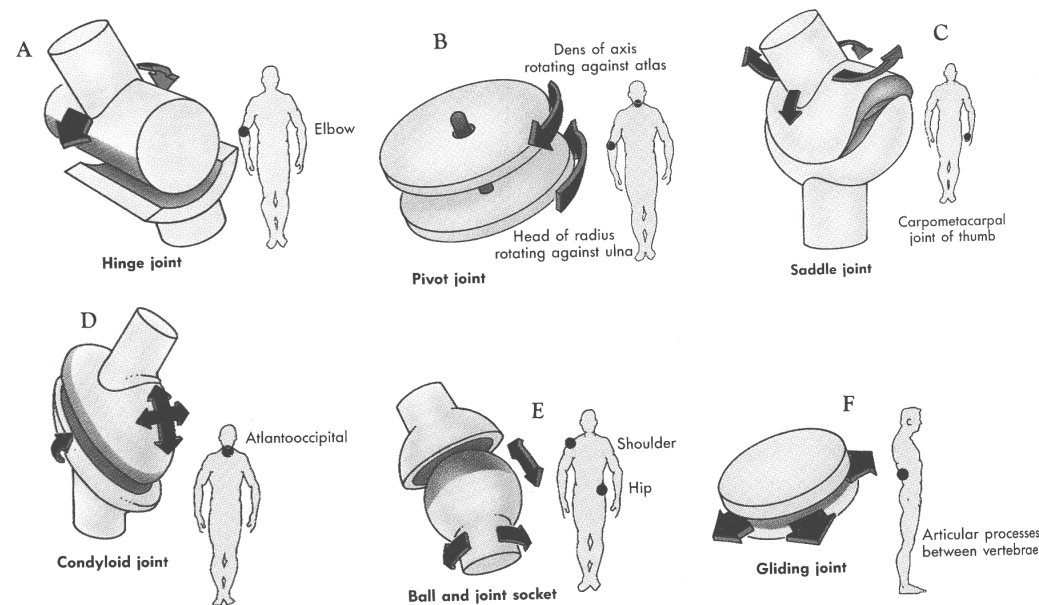
Links & Joints

Link

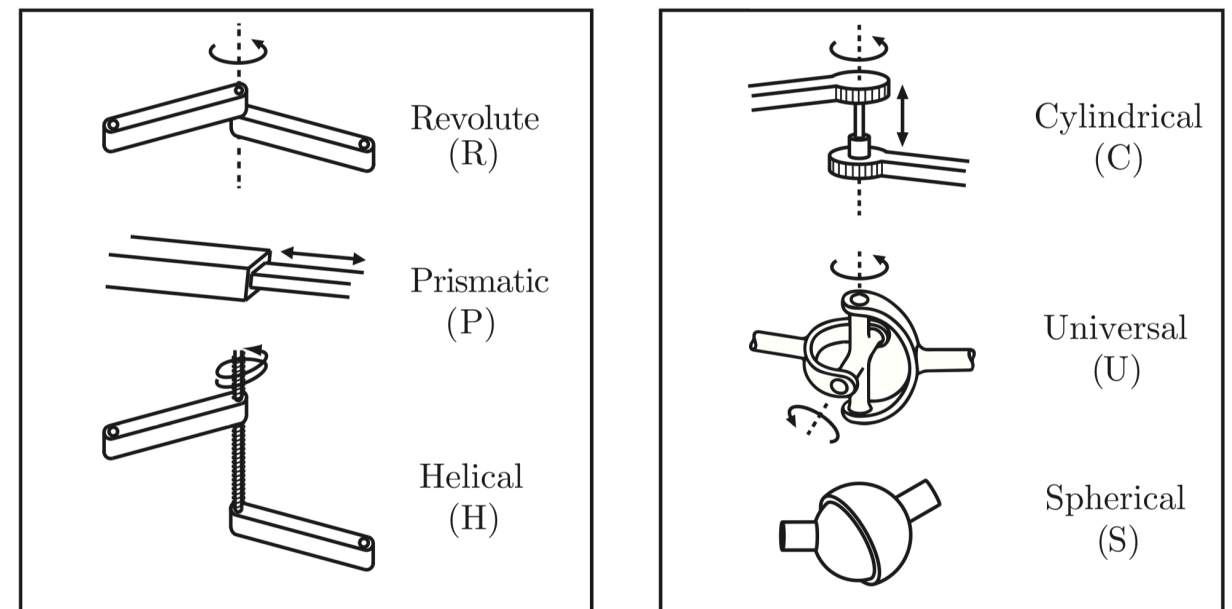
A rigid body which defines the relationship between two neighboring joint axes of the manipulator

Joint

A connection between links that allow (constrain) relative motion between them



Anatomical Joints



Typical Robot Joints

Forward Manipulator Kinematics

Lower Pair

Lower Pair

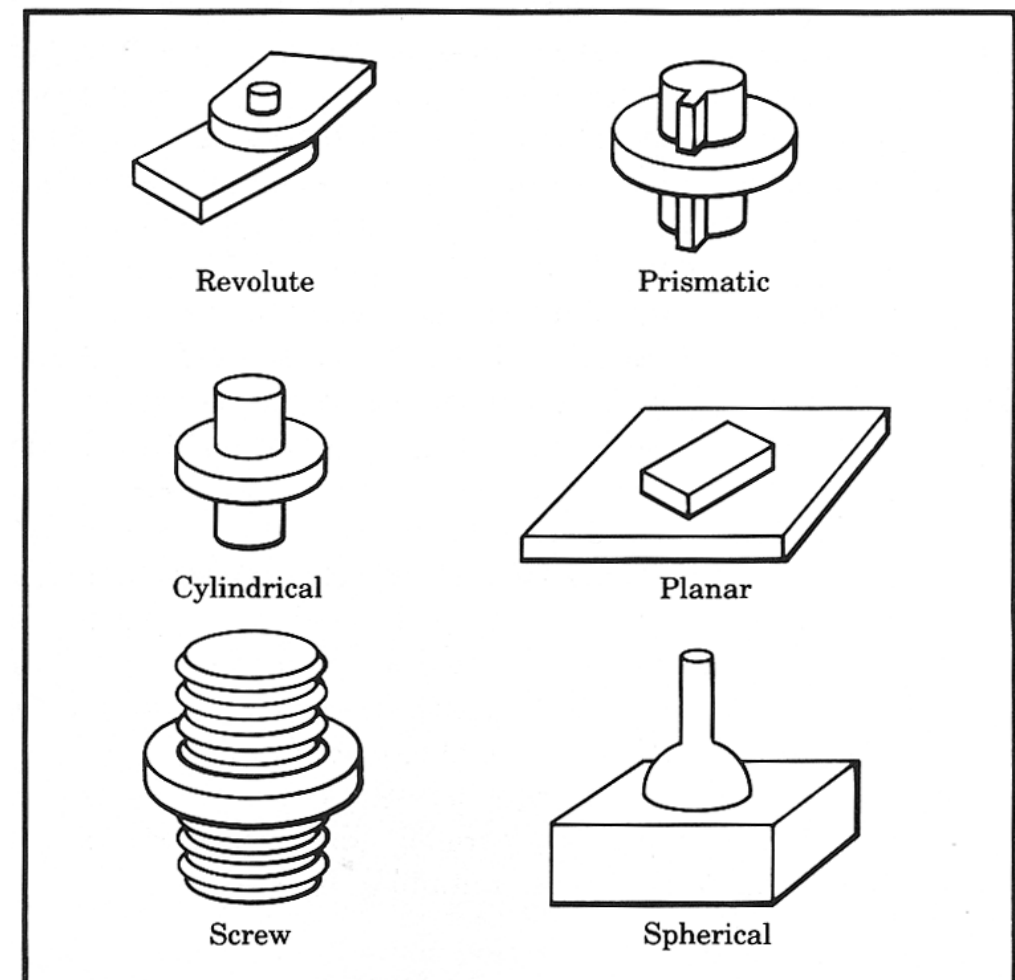
The connection between a pair of bodies (joint) when the relative motions is characterize by two *surfaces* sliding over one another (vs. Higher Pair where the contact is a line, curve or a point, i.e. cam)

Mechanical Design Constrains

For robotic manipulators 2 type of joints are mostly used:

Revolute Joint (1 DOF)

Prismatic Joint (1 DOF)



Forward Manipulator Kinematics

DH Parameters: Preview

Link Length a_{i-1}

The *distance* from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ measured along $\hat{X}_{i-1}$

Link Twist α_{i-1}

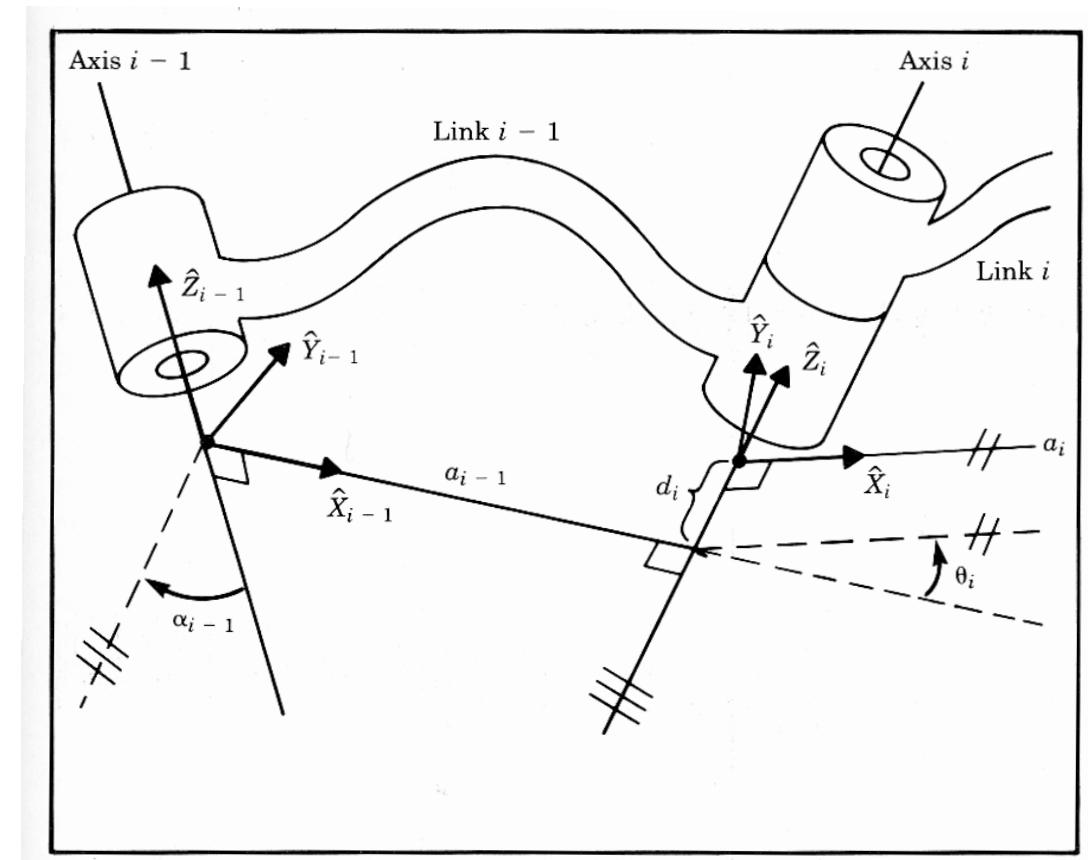
The *angle* between $\hat{Z}_{i-1}$ and $\hat{Z}_i$ measured about $\hat{X}_{i-1}$

Link Offset d_i

The *distance* from $\hat{X}_{i-1}$ to $\hat{X}_i$ measured along $\hat{Z}_i$

Link Angle θ_i

The *angle* between $\hat{X}_{i-1}$ and $\hat{X}_i$ measured about $\hat{Z}_i$



Note: $a_i \geq 0$

α_i d_i θ_i are signed quantities

Forward Manipulator Kinematics

DH Parameters: Length & Twist (Link Parameters)

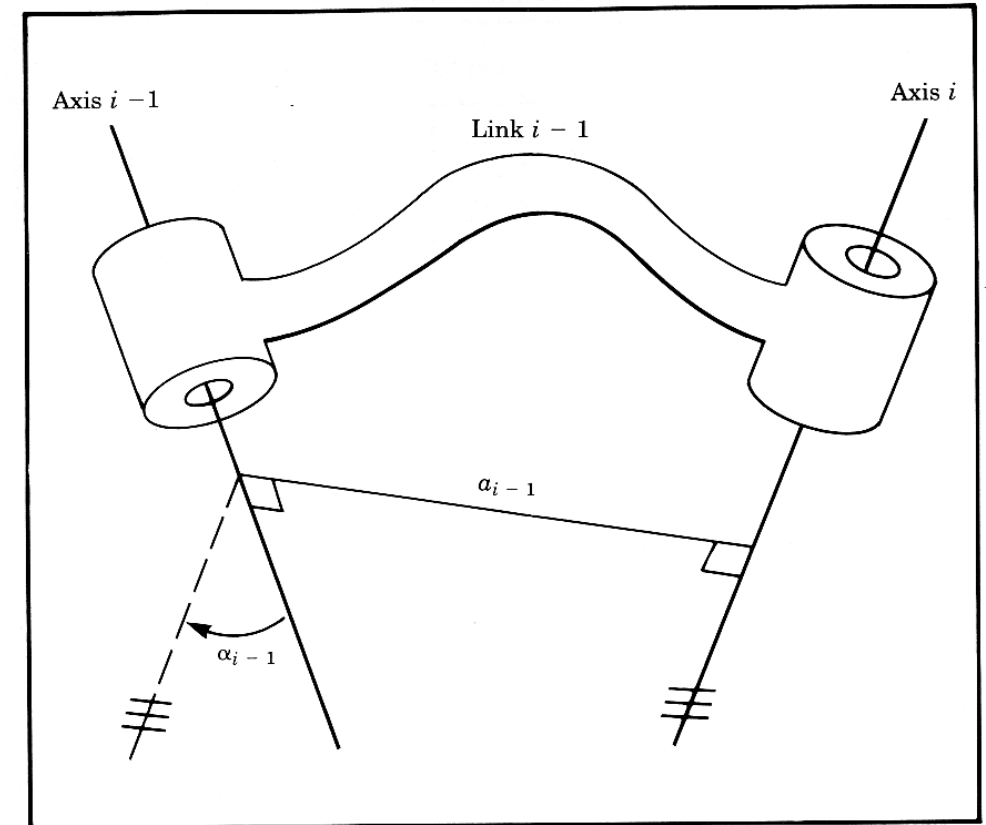
Link Length a_{i-1}

The *distance* from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ measured along $\hat{X}_{i-1}$

Link Twist α_{i-1}

The *angle* between $\hat{Z}_{i-1}$ and $\hat{Z}_i$ measured about $\hat{X}_{i-1}$

.....



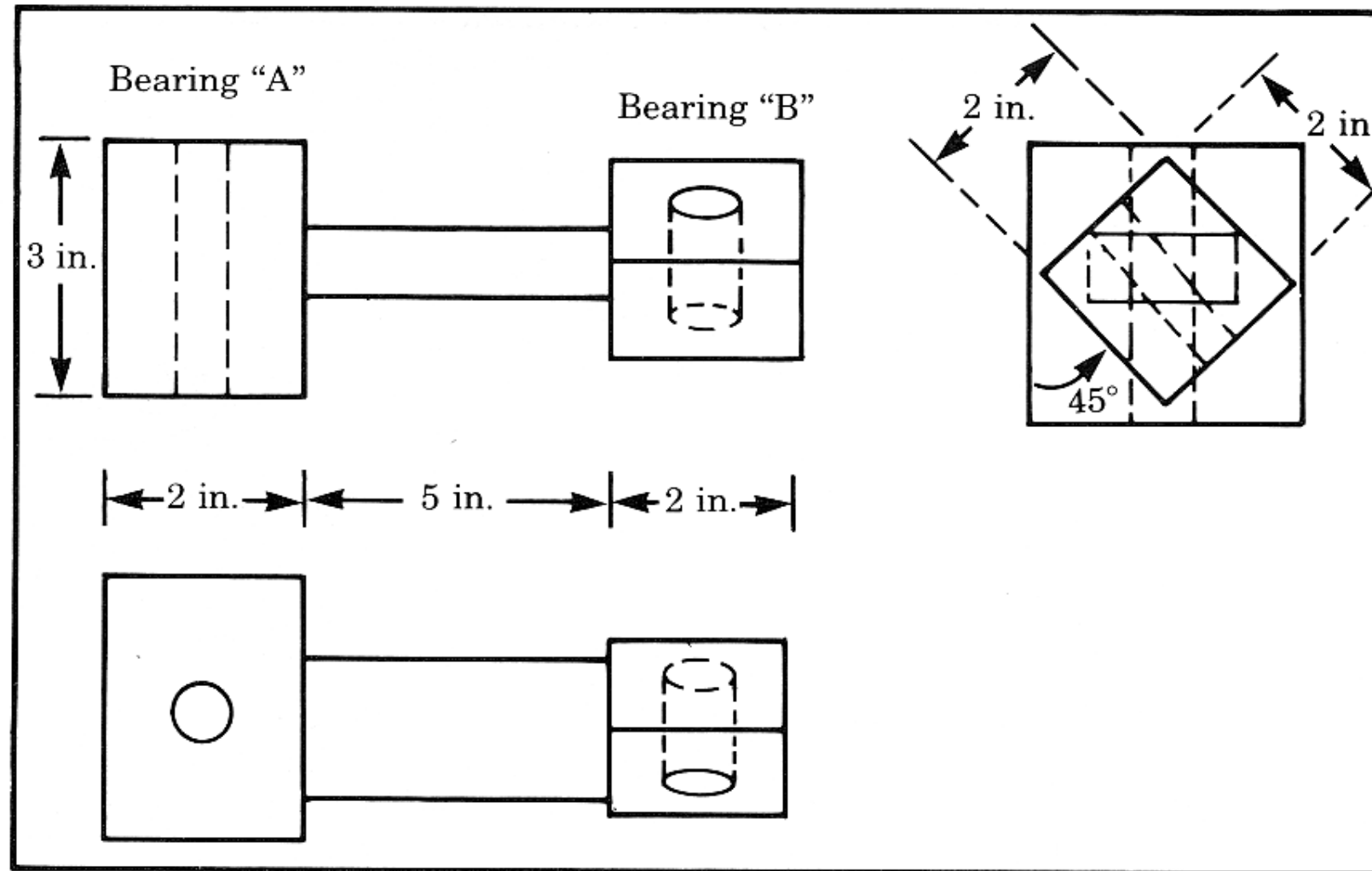
Joint Axis: A line in space (a vector with direction) about which link i rotates relative to link $i-1$

Link Length a_{i-1} : The distance between axis i and axis $i-1$
(Note: Common Normal, Sign is always positive. $a_i \geq 0$)

Link Twist α_{i-1} : The angle from axis $i-1$ to axis i measured along the common normal
(Note : Sign α_i determined by right hand rule)

Forward Manipulator Kinematics

Link Parameters: Example



Axes?

Link Length?

Link Twist?

$$a_{i-1} = 7in$$

$$\alpha_{i-1} = 45^\circ$$

Forward Manipulator Kinematics

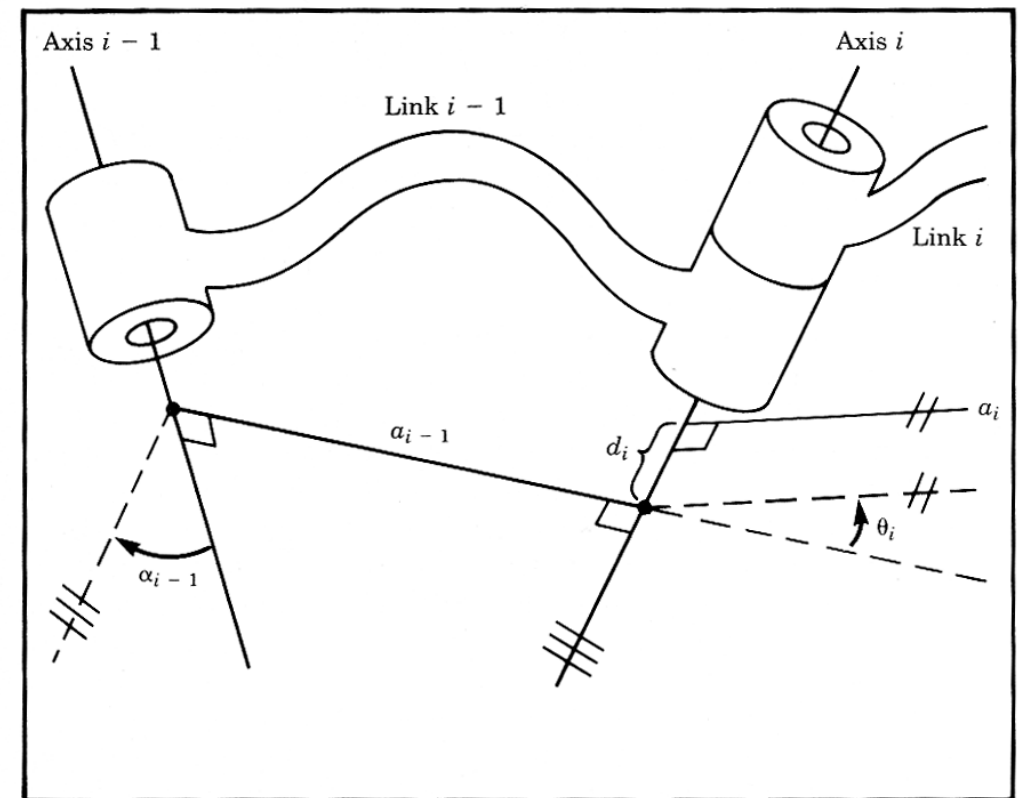
DH Parameters: Offset & Angle (Joint Variable)

Link Offset d_i

The *distance* from $\hat{X}_{i-1}$ to $\hat{X}_i$ measured along $\hat{Z}_i$

Link Angle θ_i

The *angle* between $\hat{X}_{i-1}$ and $\hat{X}_i$ measured about $\hat{Z}_i$



Forward Manipulator Kinematics

DH Parameters: Offset & Angle (Joint Variable)

Link Offset d_i :

The signed distance measured along the common axis i (joint i) from the point where a_{i-1} intersect the axis i to the point where a_i intersect the axis i

Note:

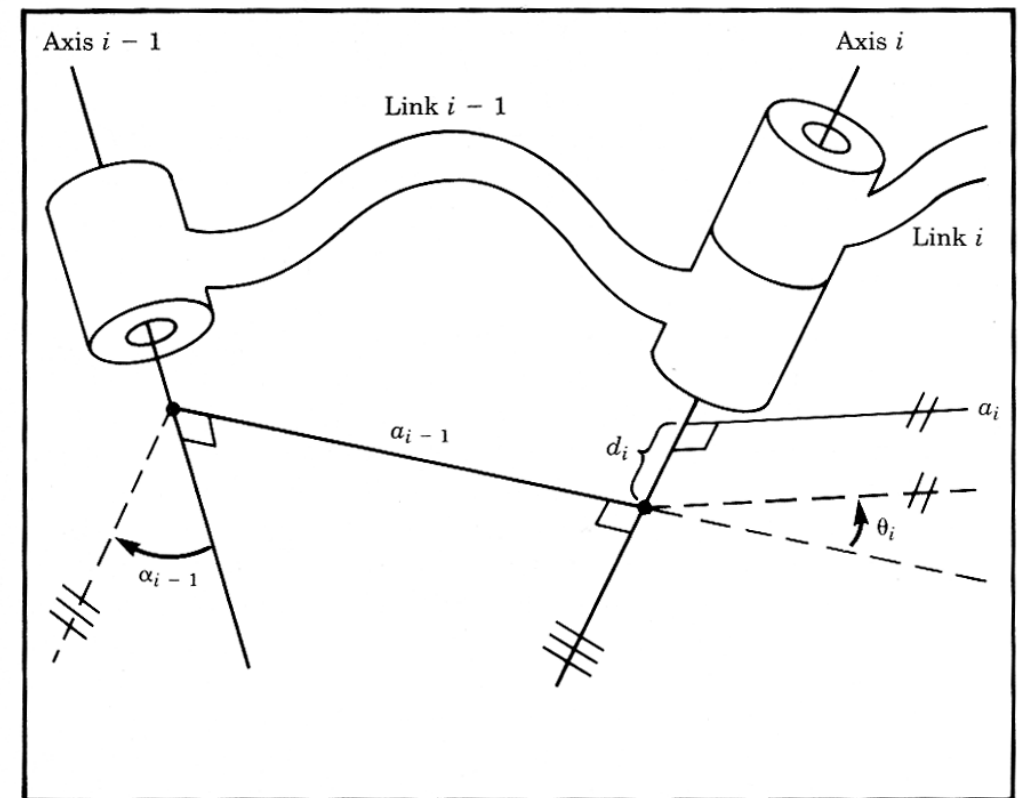
- * The link offset d_i is variable of joint i if joint i is prismatic
- * Sign of d_i depends on the choice of direction for axis i

Link Angle θ_i :

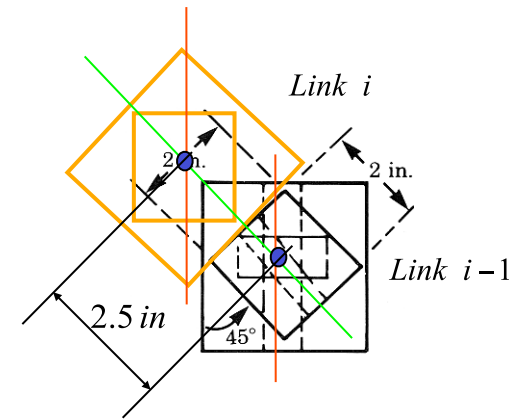
The signed angle made between an extension of a_{i-1} and a_i measured about the the axis of the joint i

Note:

- * The joint angle θ_i is variable of if the joint i is revolute
- * Sign of θ_i depends on the choice of direction for axis i



Joint Variables: Example


$$d_i = 2.5in$$

Forward Manipulator Kinematics

Affixing Frames to Links: Intermediate Links

Origin of Frame $\{i\}$

The origin of frame $\{i\}$ is located where the distance perpendicular intersects the joint i axis

Z Axis $\hat{Z}_i$

The $\hat{Z}_i$ axis of frame $\{i\}$ is coincident with the joint axis i

X Axis $\hat{X}_i$

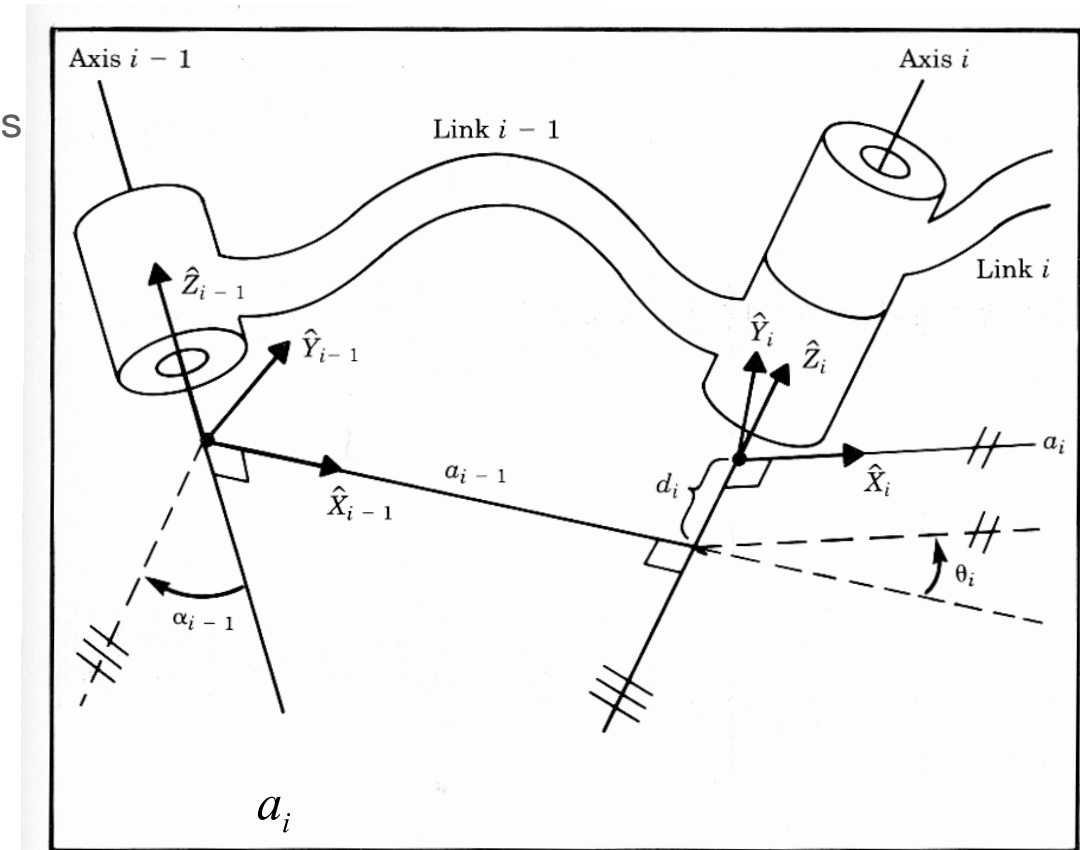
The $\hat{X}_i$ points along the distance a_i in the direction from joint i to joint $i+1$

Note:

- * For $a_i = 0$ $\hat{X}_i$ is normal to the plane of $\hat{Z}_i$ and $\hat{Z}_{i+1}$
- * The link twist angle α_i is measured in a right hand sense about $\hat{X}_i$

Y Axis $\hat{Y}_i$

The $\hat{Y}_i$ axis complete frame $\{i\}$ following the right hand rule



Forward Manipulator Kinematics

Affixing Frames to Links: First & Last Links

Frame {0}

The frame attached to the base of the robot or link 0 called frame {0} This frame does not move and for the problem of arm kinematics can be considered as the **reference frame**.

$$\begin{aligned} \text{Frame \{0\} coincides with Frame \{1\}} & \begin{cases} \alpha_0 = 0 \\ a_0 = 0 \end{cases} \\ \text{Joint 1 - Revolute Joint} & \begin{cases} \theta_1 = 0 & \text{Arbitrary} \\ d_1 = 0 & \text{Convention} \end{cases} \\ \text{Joint 1 - Prismatic Joint} & \begin{cases} \theta_1 = 0 & \text{Convention} \\ d_1 = 0 & \text{Arbitrary} \end{cases} \end{aligned}$$

$$\begin{aligned} \text{Frame \{N\} -} & \begin{aligned} & \text{Joint N - Revolute Joint} \begin{cases} \text{Align } \hat{X}_N \text{ with } \hat{X}_{N-1} \Rightarrow \theta_N = 0 \\ \{N\} \text{ Origin} \Rightarrow d_N = 0 \end{cases} \\ & \text{Joint N - Prismatic Joint} \begin{cases} \hat{X}_N \text{ Direction} \Rightarrow \theta_N = 0 \\ \{N\} \text{ Origin} - \text{Intersect of } \hat{X}_{N-1} \text{ \& joint axis } n \Rightarrow d_N = 0 \end{cases} \end{aligned} \end{aligned}$$

Forward Manipulator Kinematics

Link Frame Attachment Procedure

Joint Axes

Identify the joint axes and imagine (or draw) infinite lines along them. For the following steps below, consider two of these neighboring lines (at axes i and $i+1$)

Frame Origin

Identify the common perpendicular between them, or point of intersection. At the point of intersection, or at the point where the common perpendicular meets the i th axis, assign the link frame origin.

$\hat{Z}_i$ Axis

Assign the $\hat{Z}_i$ axis pointing along the i th joint axis.

$\hat{X}_i$ Axis

Assign the $\hat{X}_i$ axis pointing along the common perpendicular, or if the axes $\hat{Z}_{i+1}$ and $\hat{Z}_i$ intersect, assign $\hat{X}_i$ to be normal to the plane containing the two $\hat{Z}_i$ axes.

$\hat{Y}_i$ Axis

Assign the $\hat{Y}_i$ axis to complete a right hand coordinate system.

Frame {0} and Frame {1}

Assign {0} to match {1} when the first joint variable is zero. For {N} choose an origin location and $\hat{X}_N$ direction freely, but generally so as to cause as many linkage parameters as possible to be zero

Forward Manipulator Kinematics

DH Parameters: Summary

Link Length a_{i-1}

The *distance* from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ measured along $\hat{X}_{i-1}$

Link Twist α_{i-1}

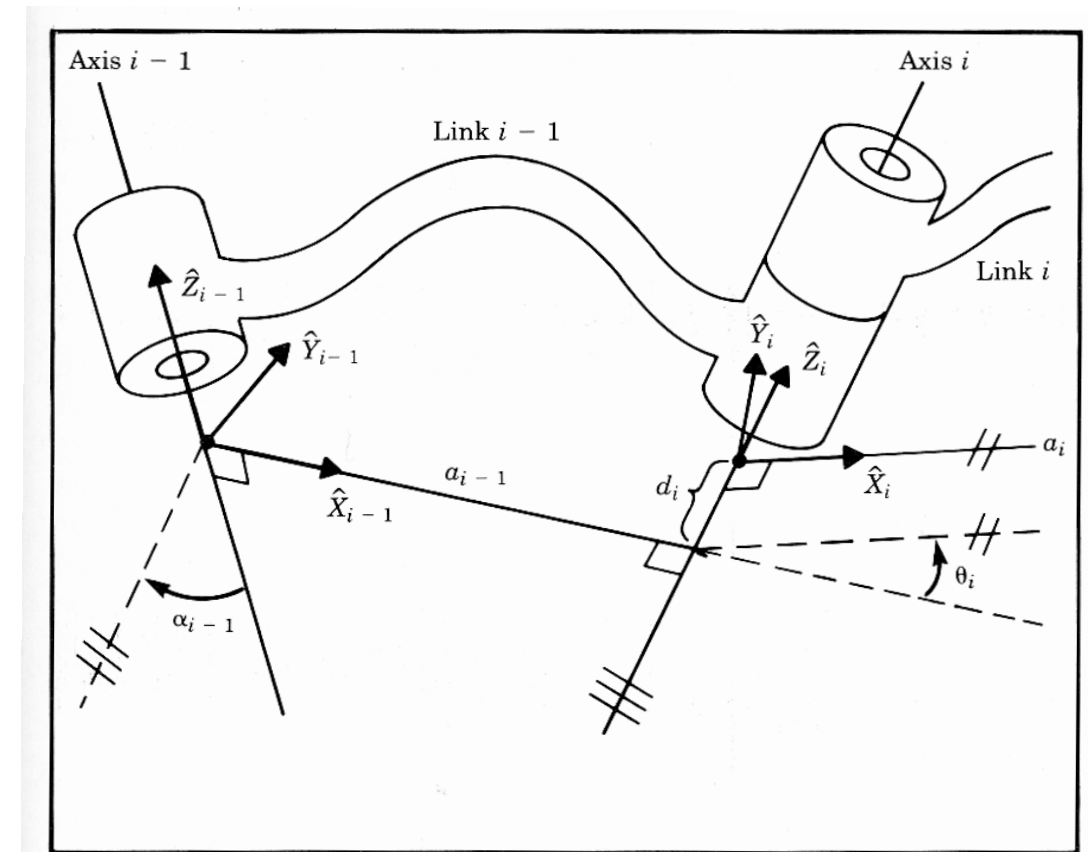
The *angle* between $\hat{Z}_{i-1}$ and $\hat{Z}_i$ measured about $\hat{X}_{i-1}$

Link Offset d_i

The *distance* from $\hat{X}_{i-1}$ to $\hat{X}_i$ measured along $\hat{Z}_i$

Link Angle θ_i

The *angle* between $\hat{X}_{i-1}$ and $\hat{X}_i$ measured about $\hat{Z}_i$

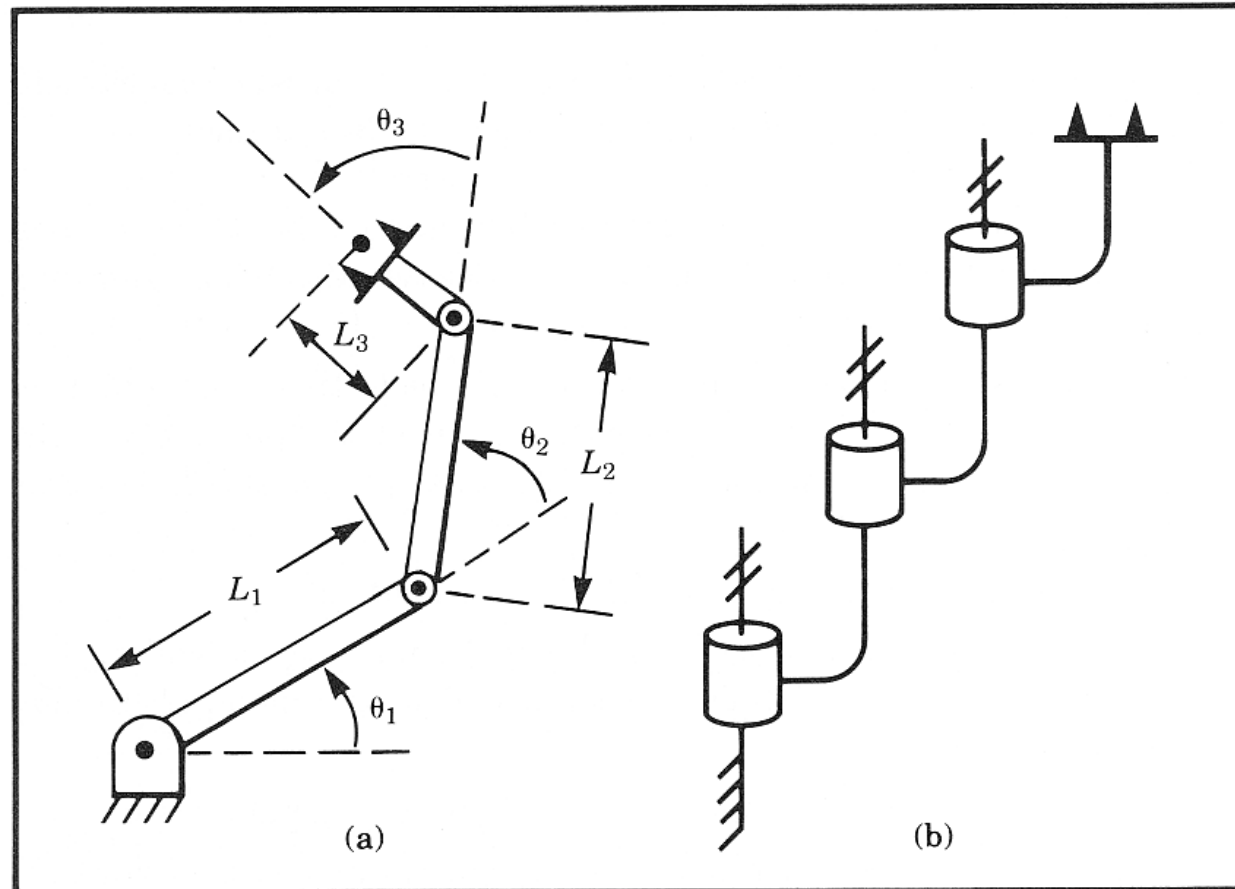


Note: $a_i \geq 0$

α_i d_i θ_i are signed quantities

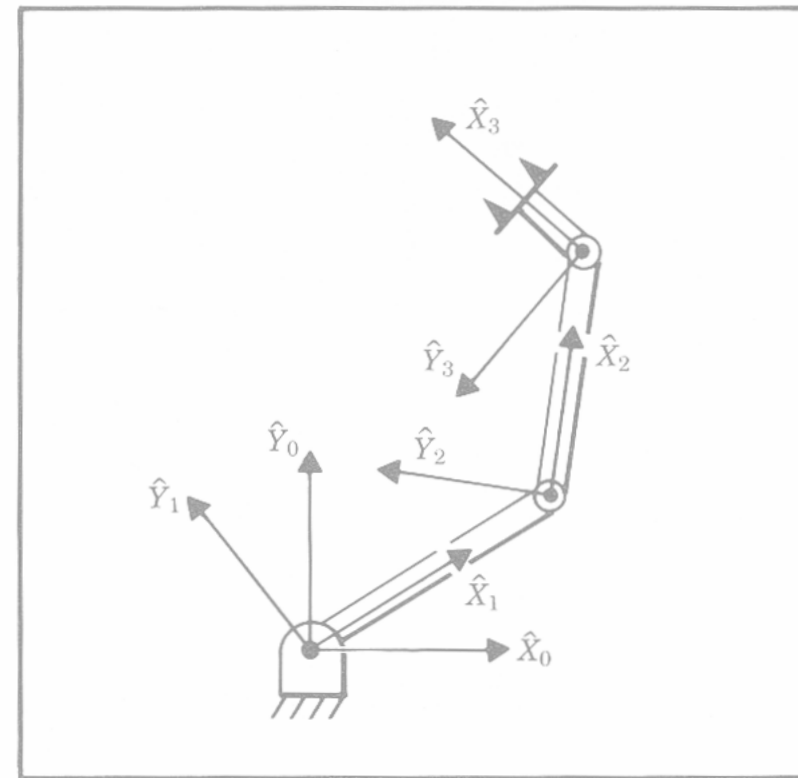
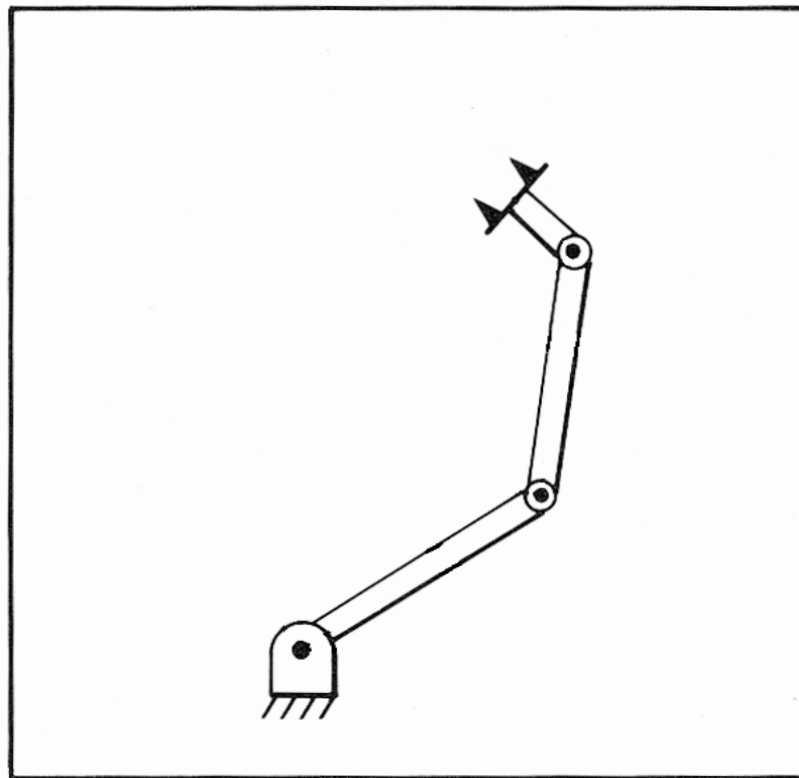
Forward Manipulator Kinematics

DH Parameters: Example 1



Forward Manipulator Kinematics

DH Parameters: Example 1

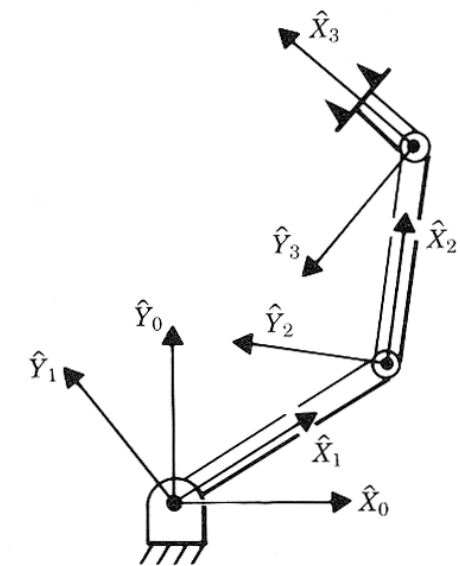
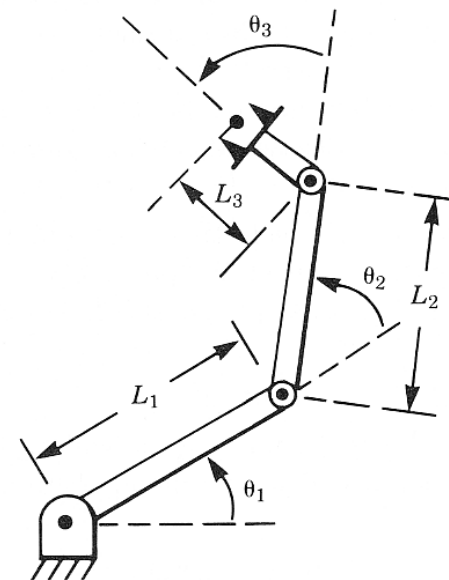


- Identify the joint axes
- Identify the common perpendicular between joint axes
- Assign the $\hat{Z}_i$ axis pointing along the i th joint axis
- Assign the $\hat{X}_i$ axis pointing along the common perpendicular
- Assign the $\hat{Y}_i$ axis to complete a right hand coordinate system
- Assign $\{0\}$ to match $\{1\}$ when the first joint variable is zero. For $\{N\}$ choose an origin location and $\hat{X}_N$ direction freely, but generally so as to cause as many linkage parameters as possible to be zero

Forward Manipulator Kinematics

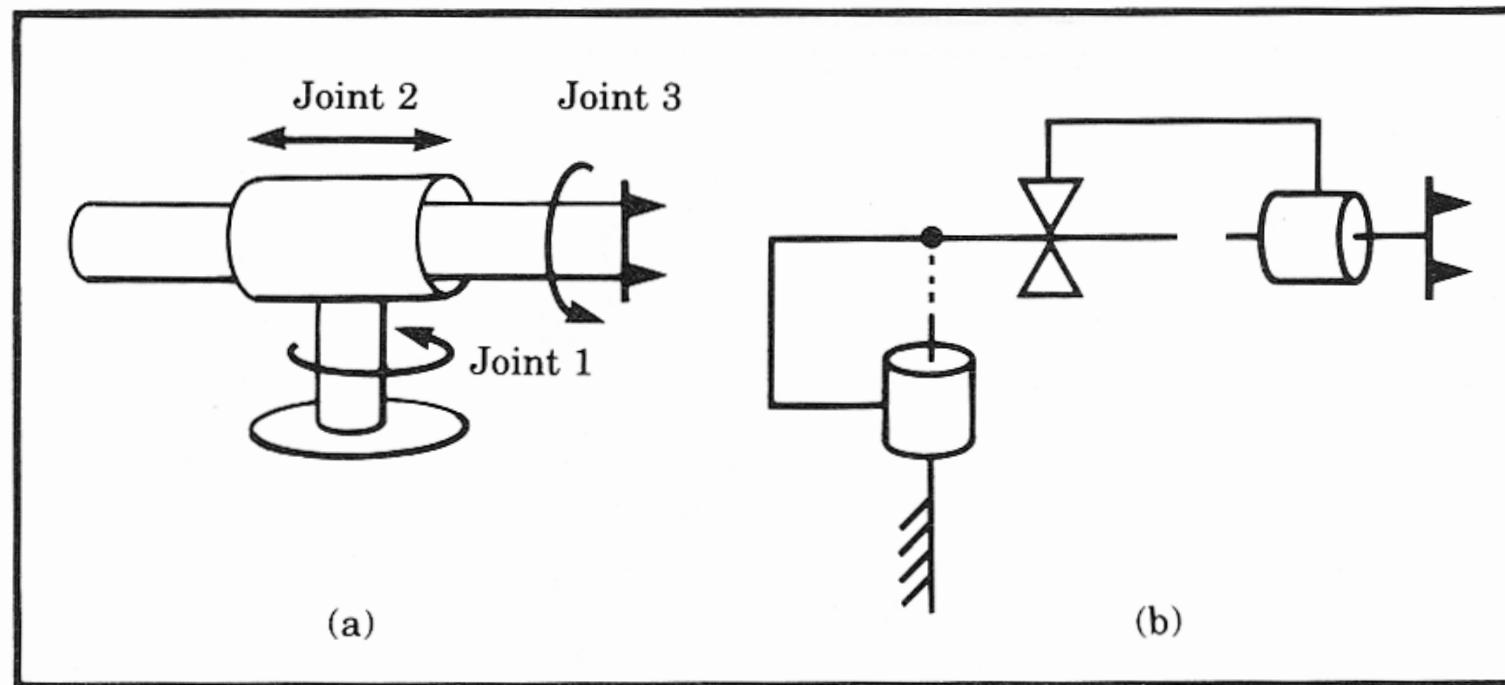
DH Parameters: Example 1

$i-1$	i	$i+1$	α_{i-1}	a_{i-1}	d_i	θ_i
0	1	2				
1	2	3				
2	3	*				



Forward Manipulator Kinematics

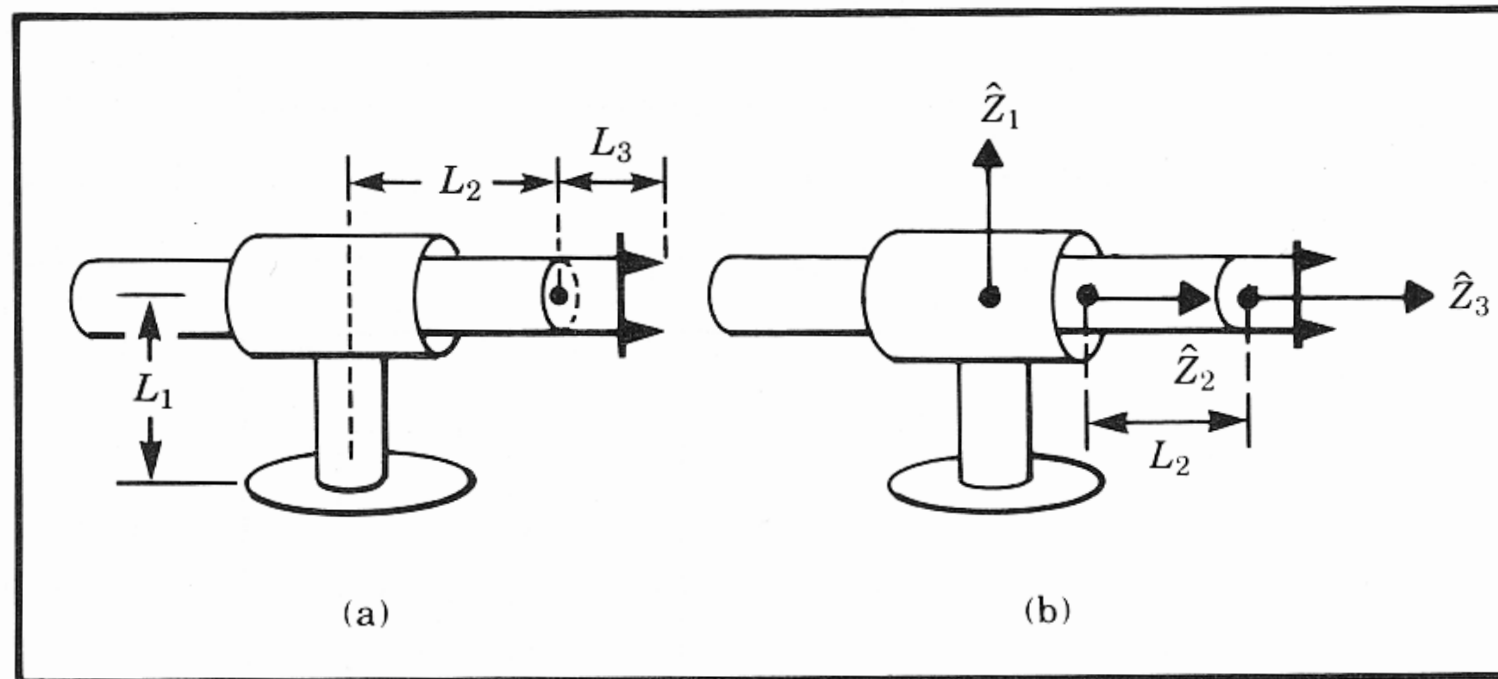
DH Parameters: Example 2



- Identify the joint axes
- Identify the common perpendicular between joint axes (?)

Forward Manipulator Kinematics

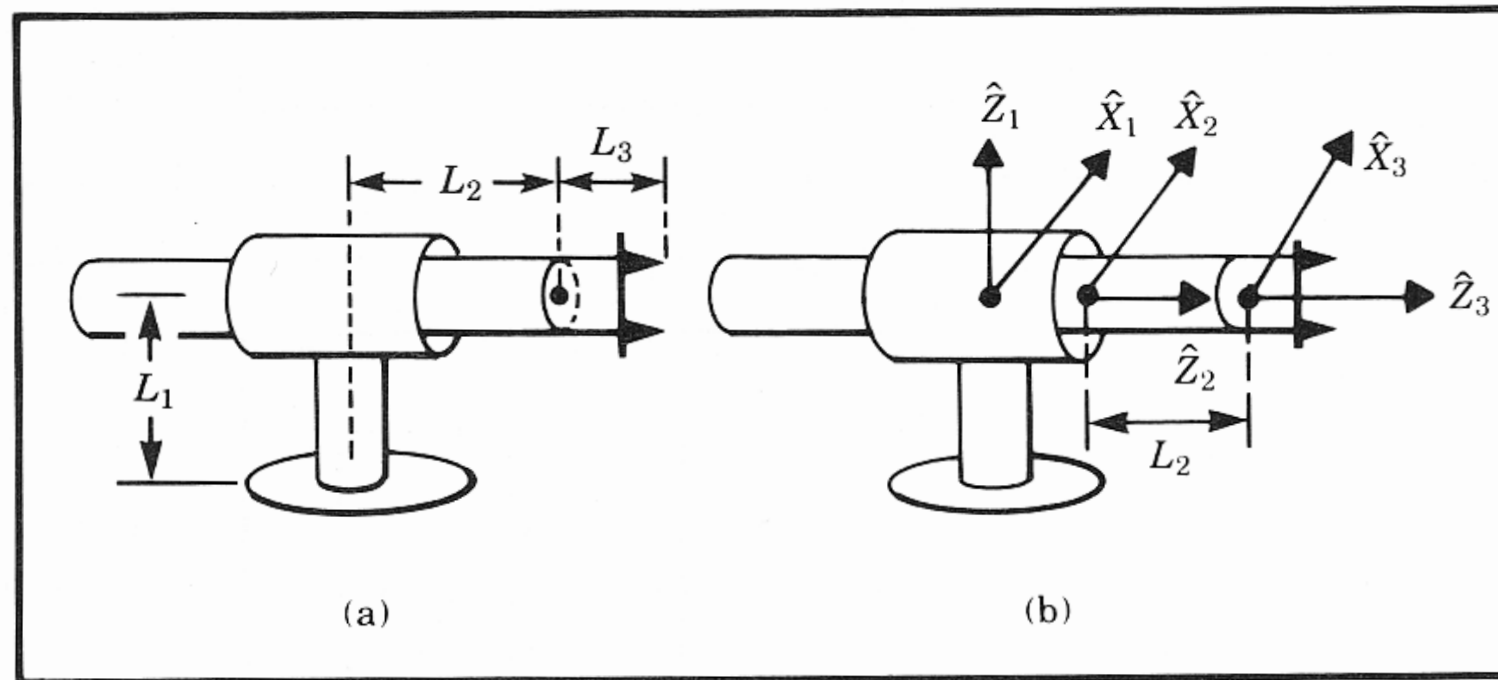
DH Parameters: Example 2



- Identify the joint axes
- Identify the common perpendicular between joint axes (?)
- Assign the $\hat{Z}_i$ axis pointing along the i th joint axis

Forward Manipulator Kinematics

DH Parameters: Example 2

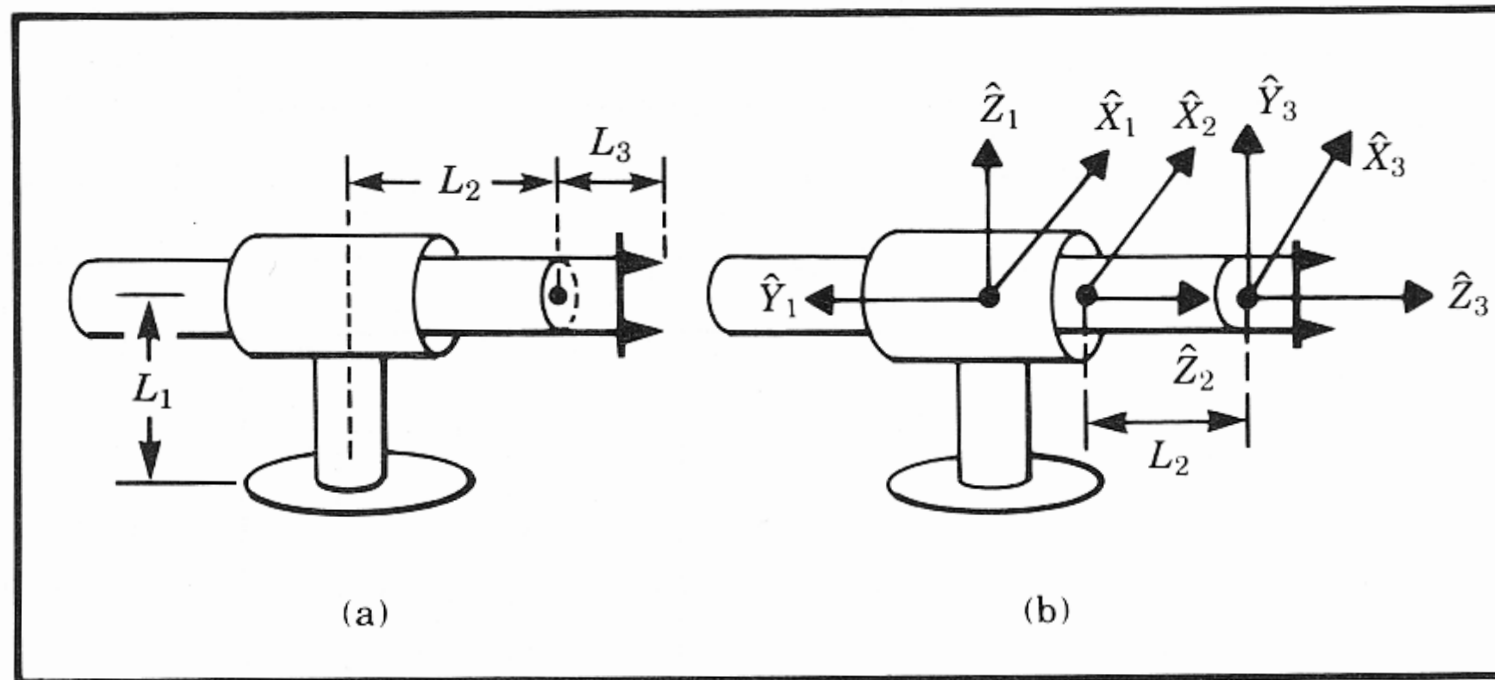


- Identify the joint axes
- Identify the common perpendicular between joint axes (?)
- Assign the $\hat{Z}_i$ axis pointing along the i th joint axis
- Assign the $\hat{X}_i$ axis pointing along the common perpendicular

if the axes $\hat{Z}_i$ and $\hat{Z}_{i+1}$ intersect, assign $\hat{X}_i$ to be normal to the plane containing the two axes

Forward Manipulator Kinematics

DH Parameters: Example 2

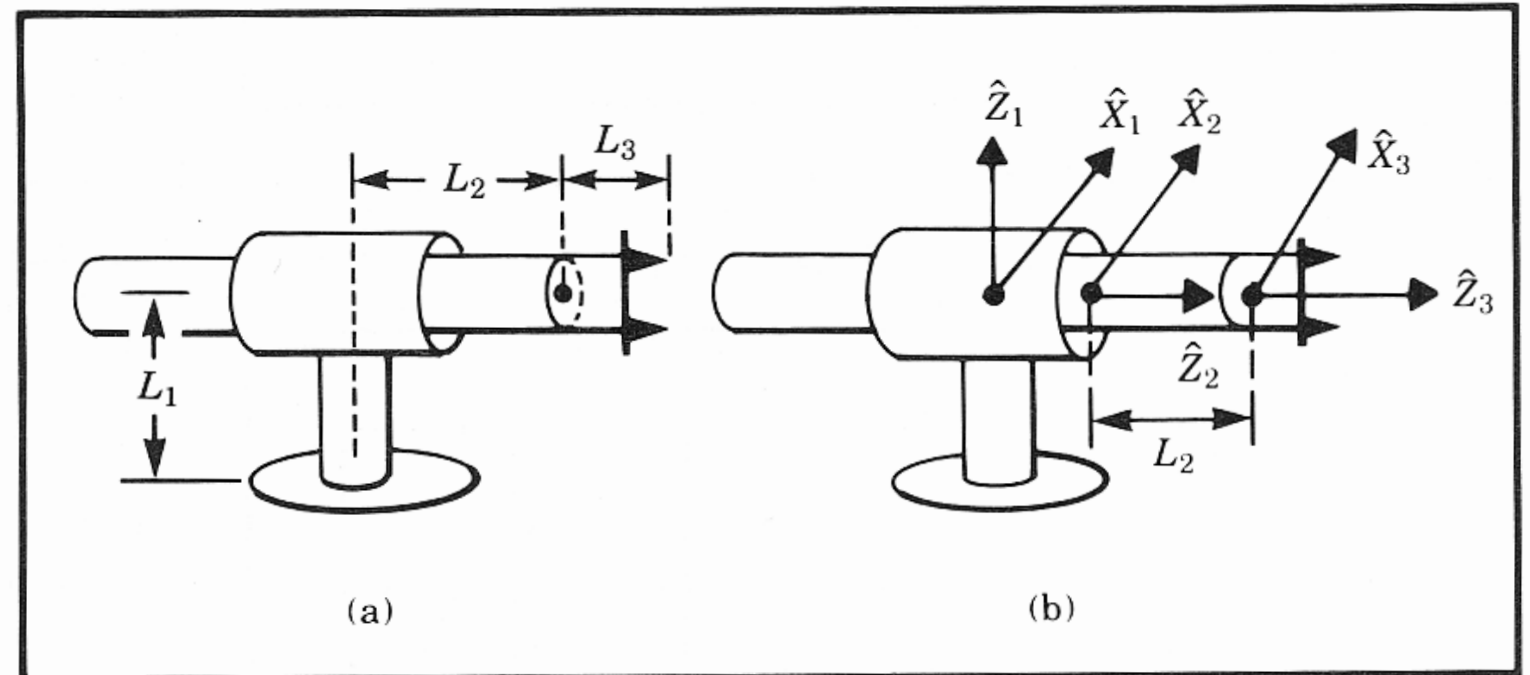


- Identify the joint axes
 - Identify the common perpendicular between joint axes (?)
 - Assign the $\hat{Z}_i$ axis pointing along the i th joint axis
 - Assign the $\hat{X}_i$ axis pointing along the common perpendicular
 - Assign the $\hat{Y}_i$ axis to complete a right hand coordinate system
 - Assign $\{0\}$ to match $\{1\}$ when the first joint variable is zero. For $\{N\}$ choose an origin location and $\hat{X}_N$ direction freely, but generally so as to cause as many linkage parameters as possible to be zero
- if the axes $\hat{Z}_i$ and $\hat{Z}_{i+1}$ intersect, assign $\hat{X}_i$ to be normal to the plane containing the two axes

Forward Manipulator Kinematics

DH Parameters: Example 2

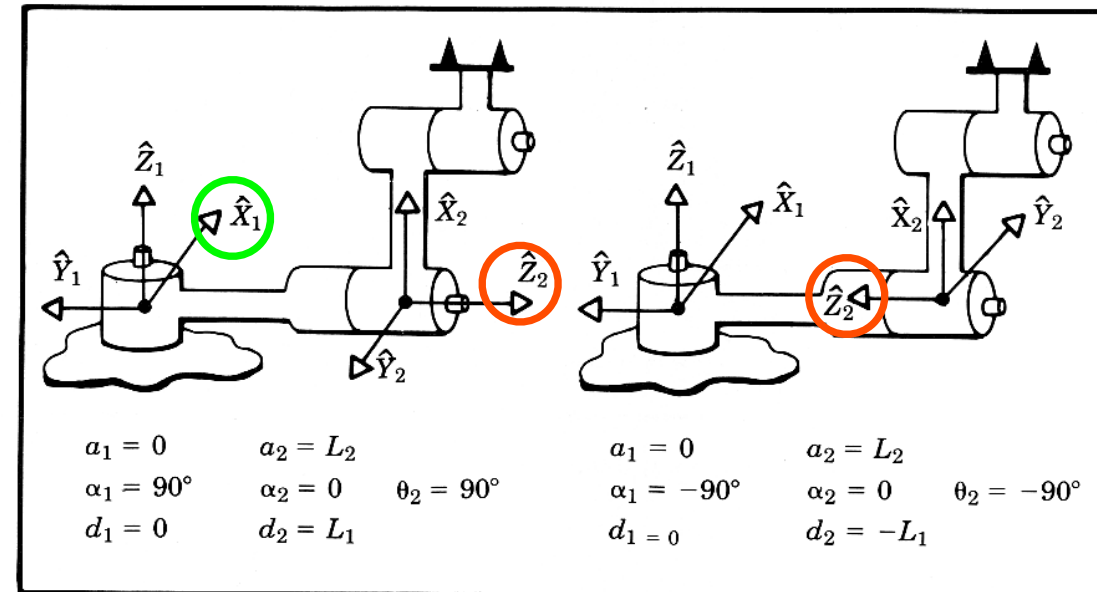
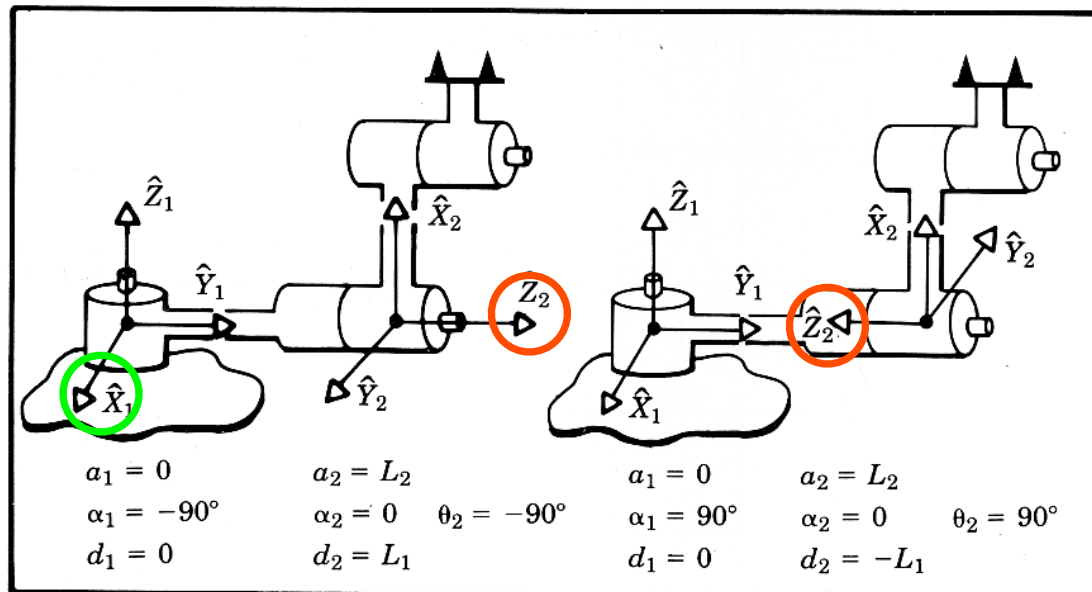
$i-1$	i	$i+1$	α_{i-1}	a_{i-1}	d_i	θ_i
0	1	2				
1	2	3				
2	3	*				



Forward Manipulator Kinematics

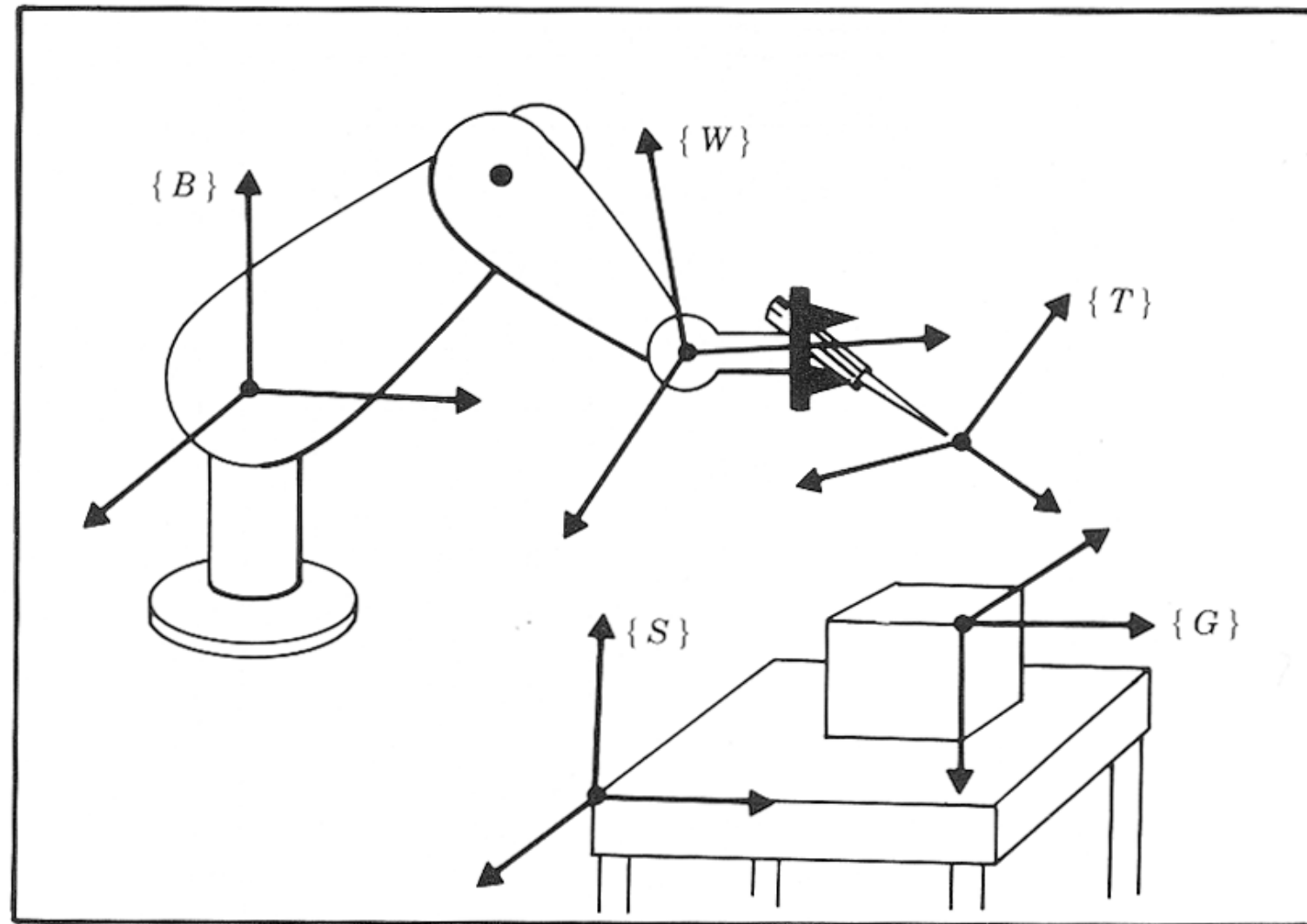
DH Parameters: Example 3

- Orthogonal Axes (Intersection) & Parallel Axes - Non Uniqueness of DH Parameters
- When $\hat{Z}_i$ and $\hat{Z}_{i+1}$ intersect there are two choices for $\hat{X}_i$ (Normal to the plane containing $\hat{Z}_i$ and $\hat{Z}_{i+1}$), as well as two choices for $\hat{Z}_{i+1}$
- There are four more possibilities corresponding to the four configuration but with $\hat{Z}_i$ pointing downward



Forward Manipulator Kinematics

Central Topic



Forward Manipulator Kinematics

DH Parameters: Review

Link Length a_{i-1}

The *distance* from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ measured along $\hat{X}_{i-1}$

Link Twist α_{i-1}

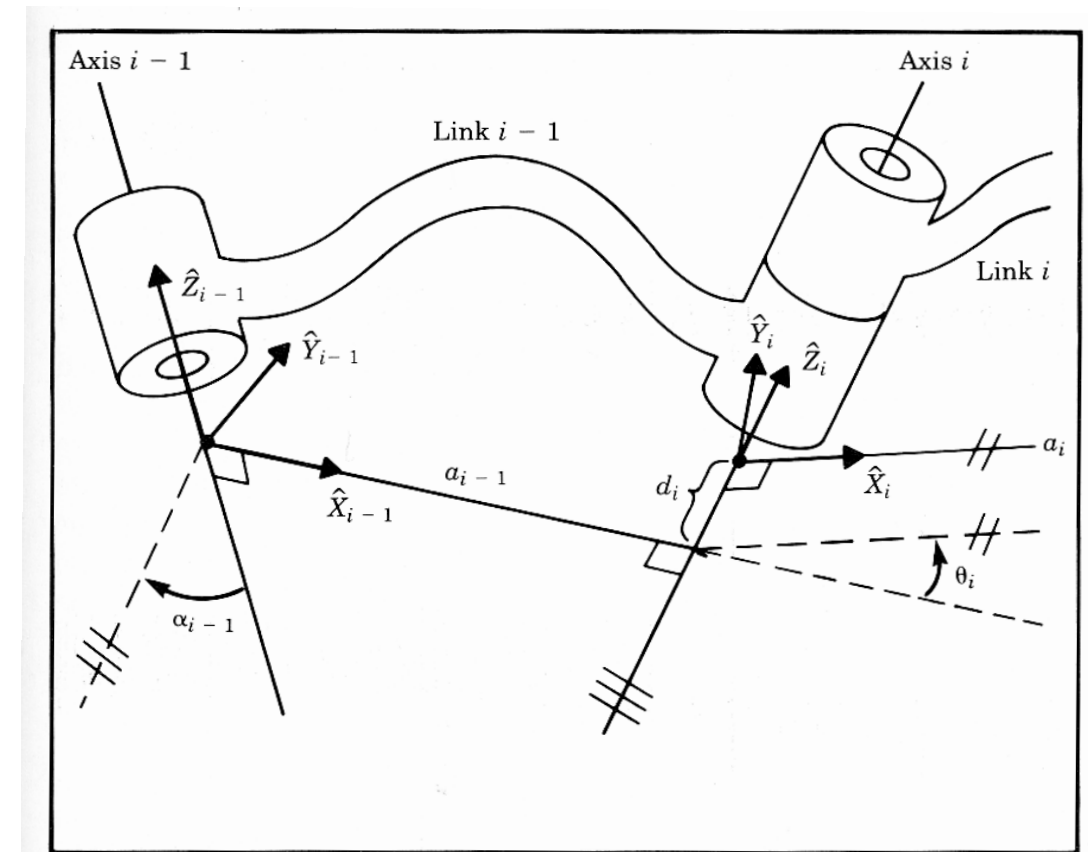
The *angle* between $\hat{Z}_{i-1}$ and $\hat{Z}_i$ measured about $\hat{X}_{i-1}$

Link Offset d_i

The *distance* from $\hat{X}_{i-1}$ to $\hat{X}_i$ measured along $\hat{Z}_i$

Link Angle θ_i

The *angle* between $\hat{X}_{i-1}$ and $\hat{X}_i$ measured about $\hat{Z}_i$



Note: $a_i \geq 0$

α_i d_i θ_i are signed quantities

Forward Manipulator Kinematics

Link Transformations

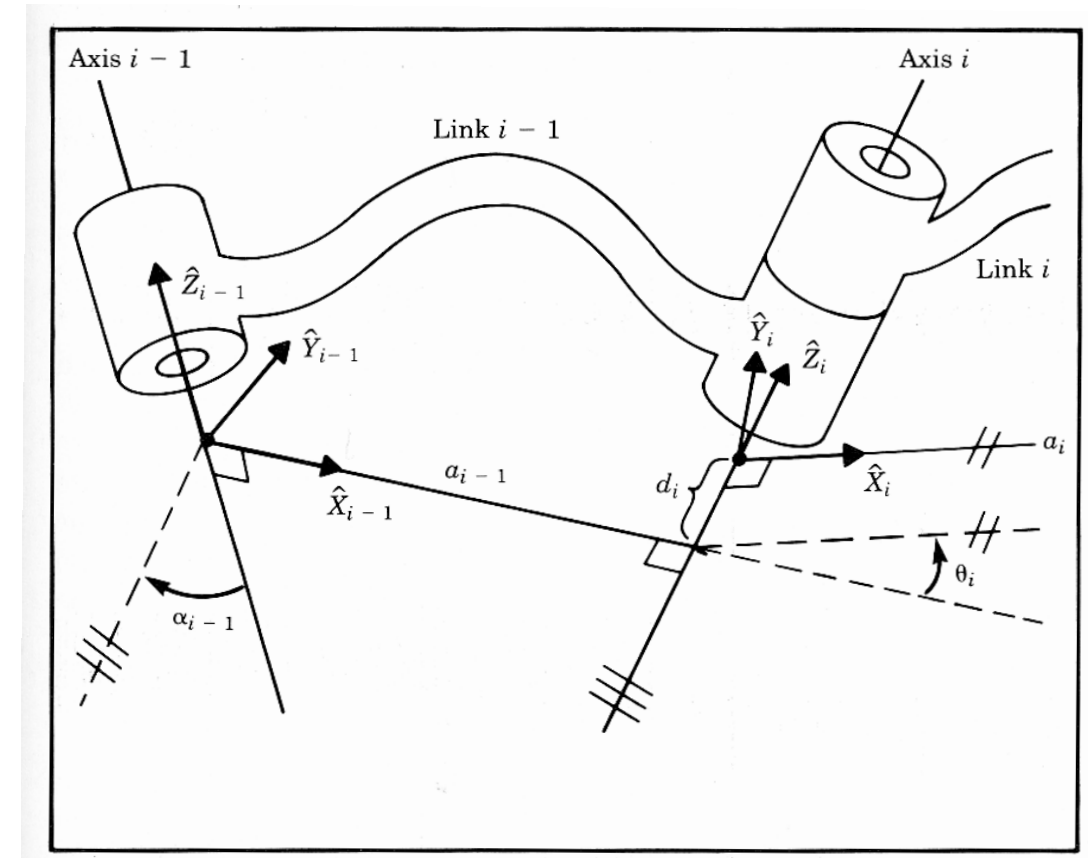
Problem: Determine the transformation which defines frame $\{i\}$ relative to the frame $\{i-1\}$

$${}^{i-1}T_i$$

Note: For any given link of a robot, ${}^{i-1}T_i$ will be a function of only one variable out of a_{i-1} α_{i-1} d_i θ_i . The other three parameters being fixed by mechanical design.

Revolute Joint $\rightarrow \theta_i$

Prismatic Joint $\rightarrow d_i$

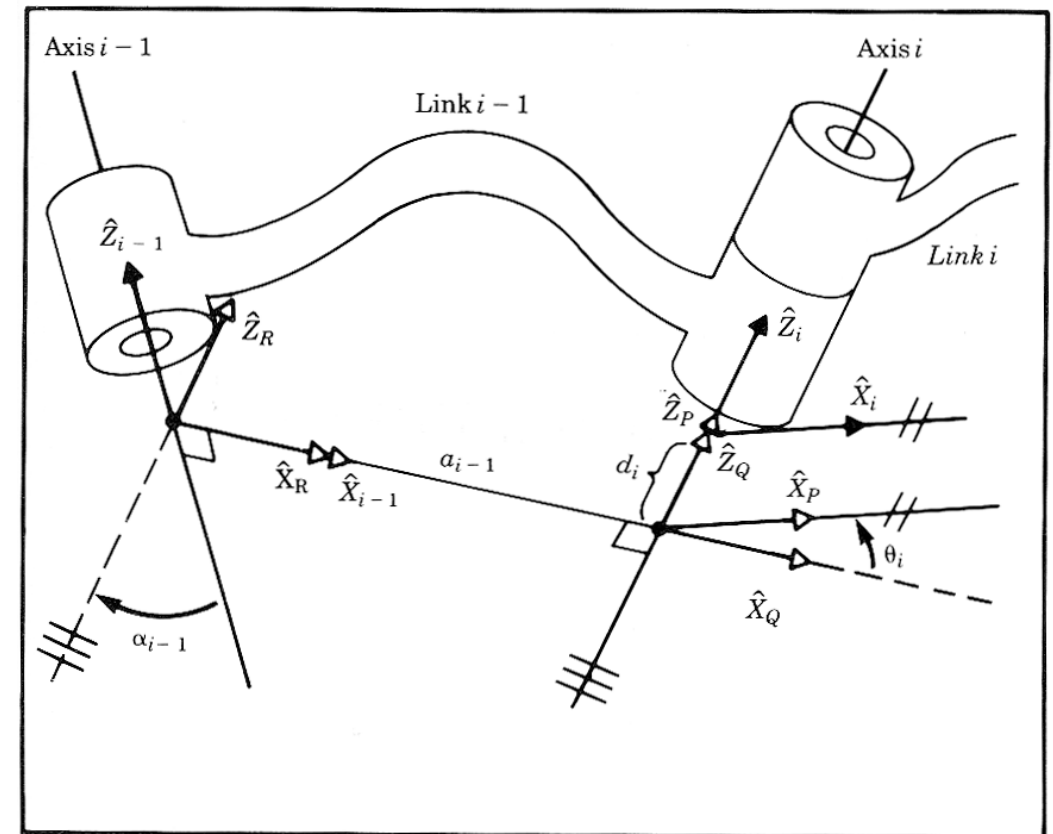


Forward Manipulator Kinematics

Link Transformations

Solution: The problem is further broken into 4 sub problem such that each of the transformations will be a function of one link parameters only

- Define three intermediate frames:
 $\{P\}$, $\{Q\}$, and $\{R\}$
 - Frame $\{R\}$ is different from $\{i-1\}$ only by a rotation of α_{i-1}
 - Frame $\{Q\}$ is different from $\{R\}$ only by a translation a_{i-1}
 - Frame $\{P\}$ is different from $\{Q\}$ only by a rotation θ_i
 - Frame $\{i\}$ is different from $\{P\}$ only by a translation d_i



Forward Manipulator Kinematics

Link Transformations

Solution: A vector defined in frame $\{i\}$ is expressed in $\{i-1\}$ as follows

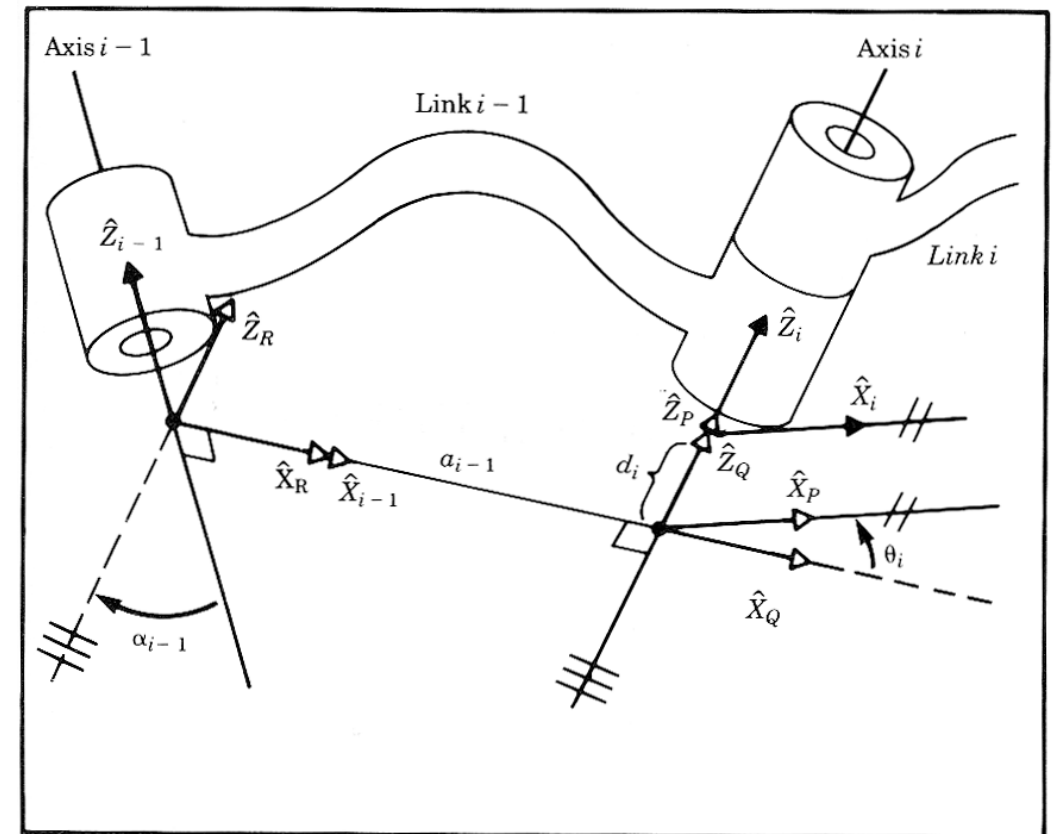
$${}^{i-1}P = {}^{i-1}T_R {}^R T_Q {}^Q T_P {}^P T_i P$$

$${}^{i-1}P = {}^{i-1}T_i P$$

The transformation from frame $\{i-1\}$ to frame $\{i\}$ is defined as follows

$${}^{i-1}T_i = {}^{i-1}T_R {}^R T_Q {}^Q T_P {}^P T_i$$

$${}^{i-1}T_i = R_X(\alpha_{i-1})D_X(a_{i-1})R_Z(\theta_i)D_Z(d_i)$$



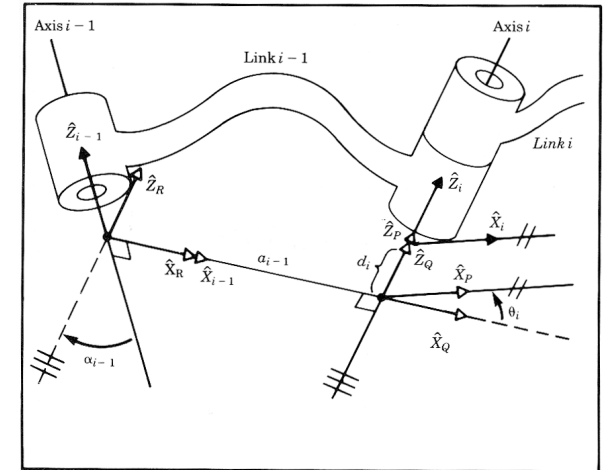
Forward Manipulator Kinematics

Link Transformations

$${}^{i-1}_iT = R_X(\alpha_{i-1})D_X(a_{i-1})R_Z(\theta_i)D_Z(d_i)$$

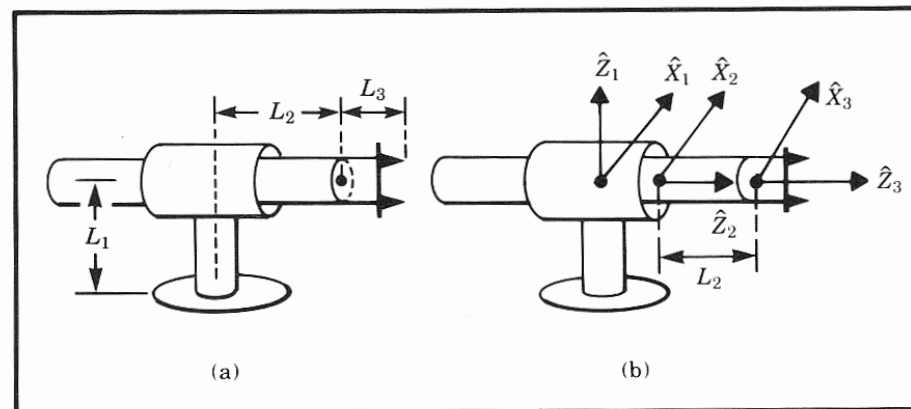
$${}^{i-1}_iT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\alpha_{i-1} & -s\alpha_{i-1} & 0 \\ 0 & s\alpha_{i-1} & c\alpha_{i-1} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_{i-1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Forward Manipulator Kinematics

Link Transformations Example



$${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	90°	0	d_2	0
3	0	0	L_2	θ_3

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -d_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & l_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

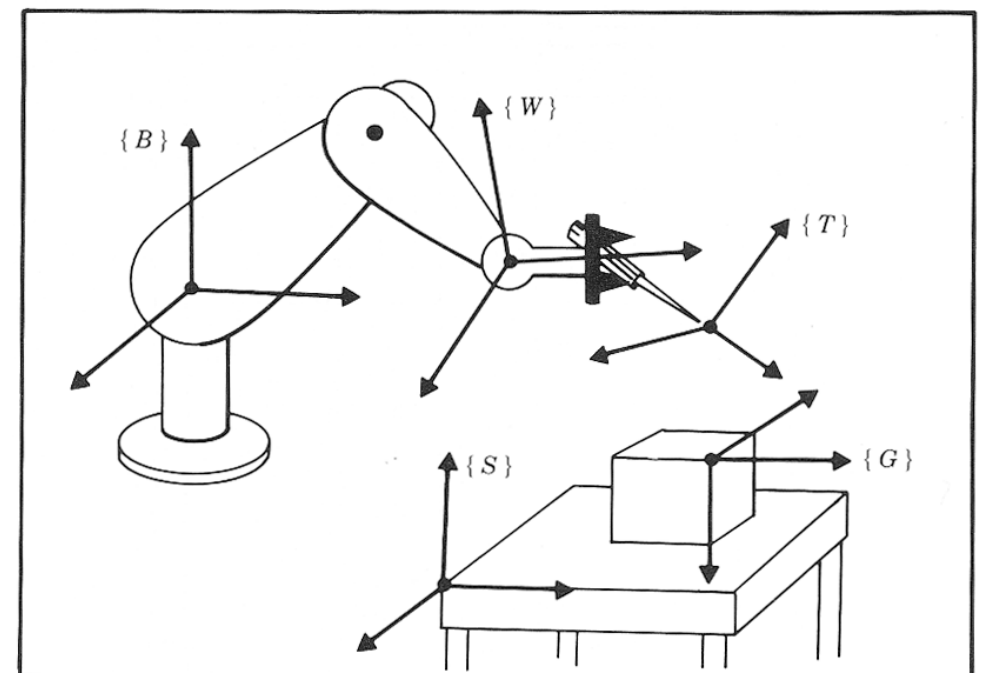
Forward Manipulator Kinematics

Concatenating Link Transformations

- Define link frames
- Define DH parameters of each link
- Compute the individual link transformation matrix
- Multiply together to find the single transformation relating frame $\{N\}$ to frame $\{0\}$

$${}^0_N T = {}^0_1 T {}^1_2 T {}^2_3 T \dots {}^{N-1}_N T$$

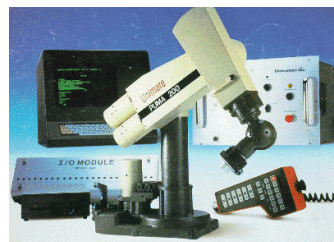
- The transformation ${}^0_N T$ will be a function of all n joint variables.
- If the robot's joint position sensors are measured, the Cartesian position and orientation of the last link may be computed by ${}^0_N T$



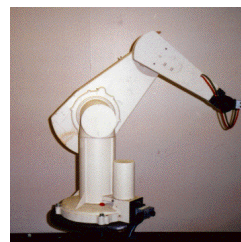
Forward Manipulator Kinematics

Example: PUMA Family

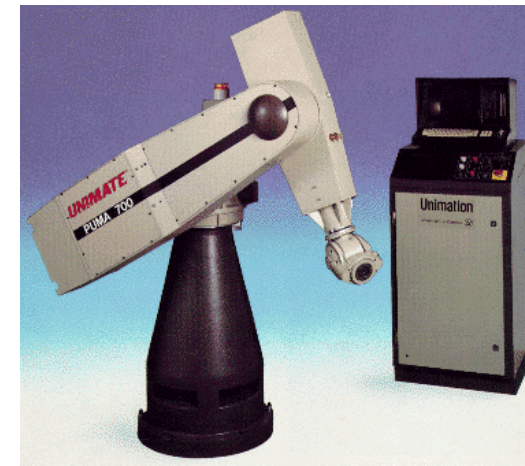
PUMA 200



PUMA 500

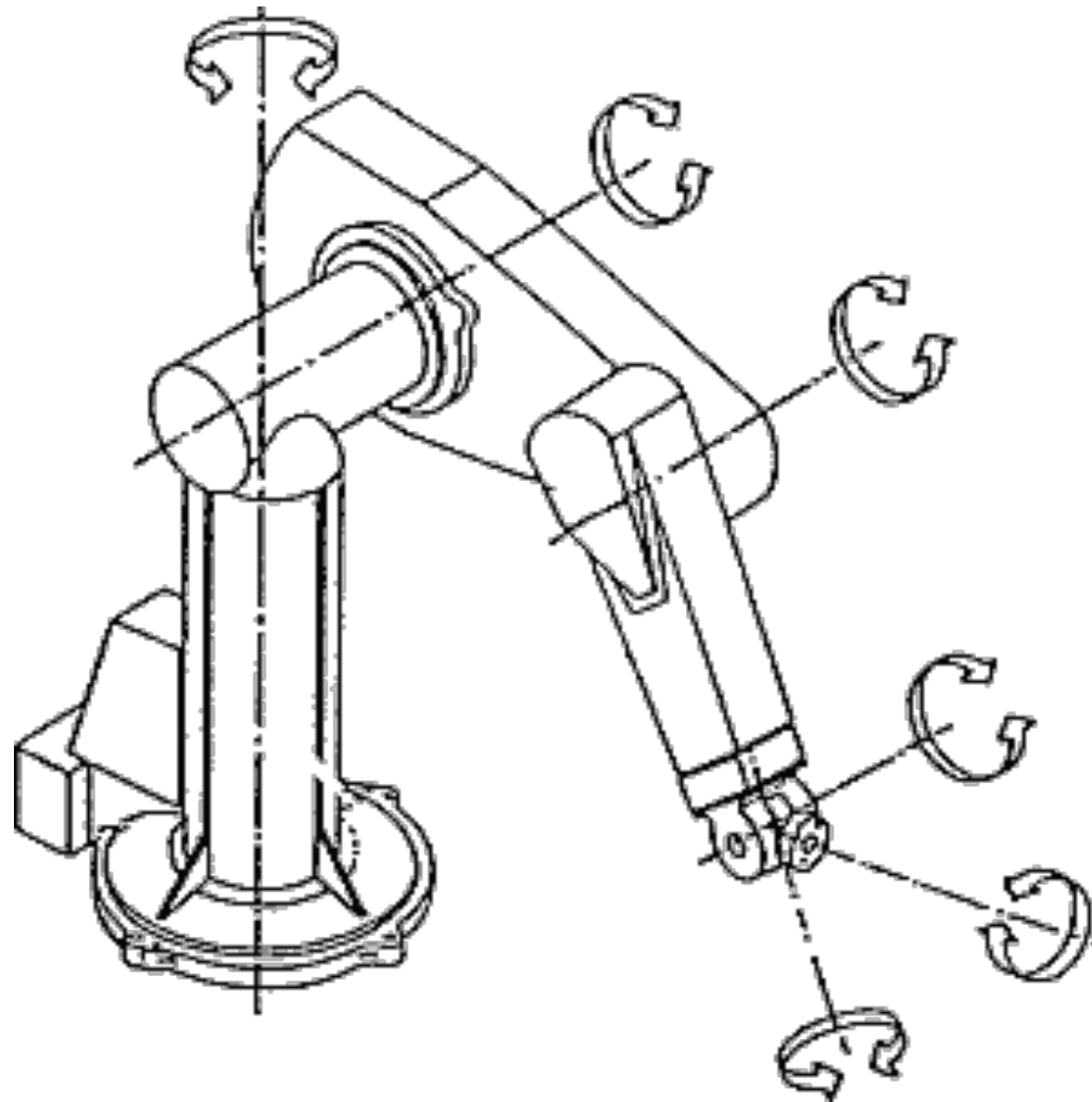


PUMA 700



Forward Manipulator Kinematics

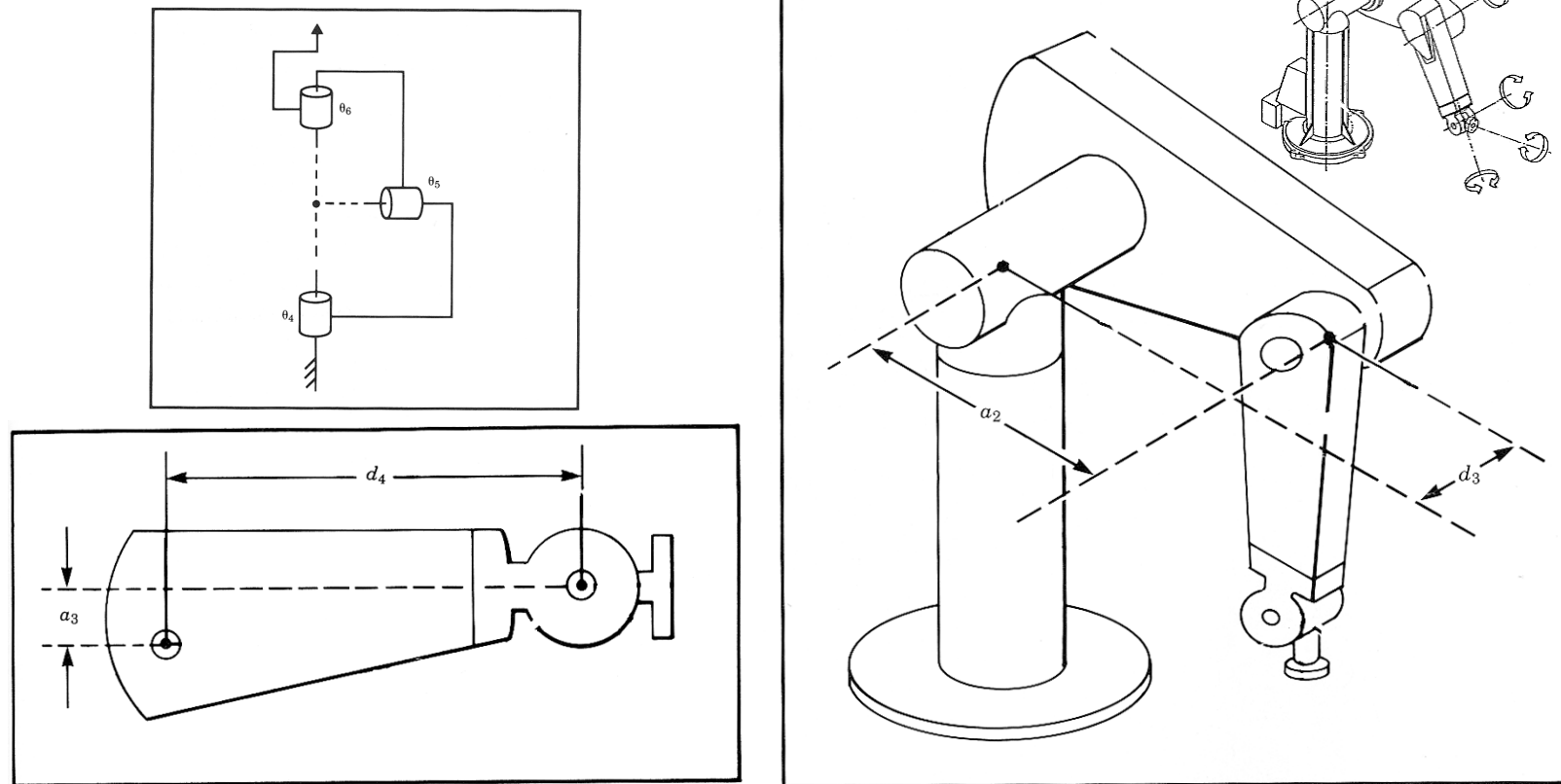
Example: PUMA 560



Forward Manipulator Kinematics

Example: Kinematics of PUMA 560

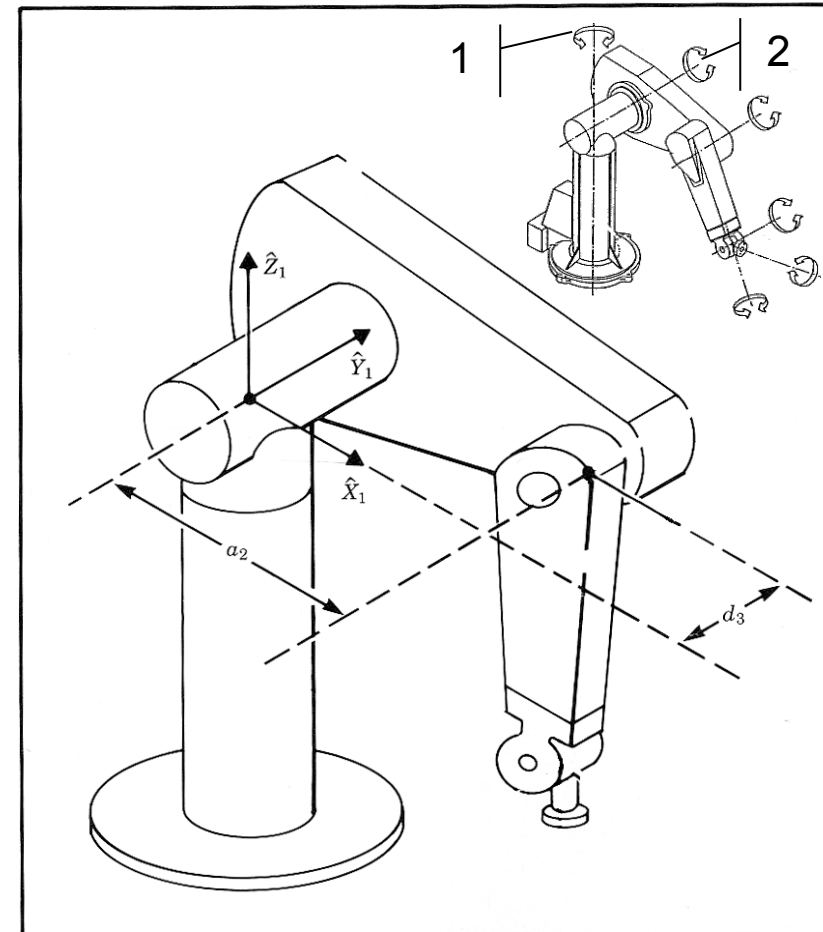
- The robot position in which all joint angles are equal to zero



Forward Manipulator Kinematics

Example: Frame Assignments - Frame {0} {1}

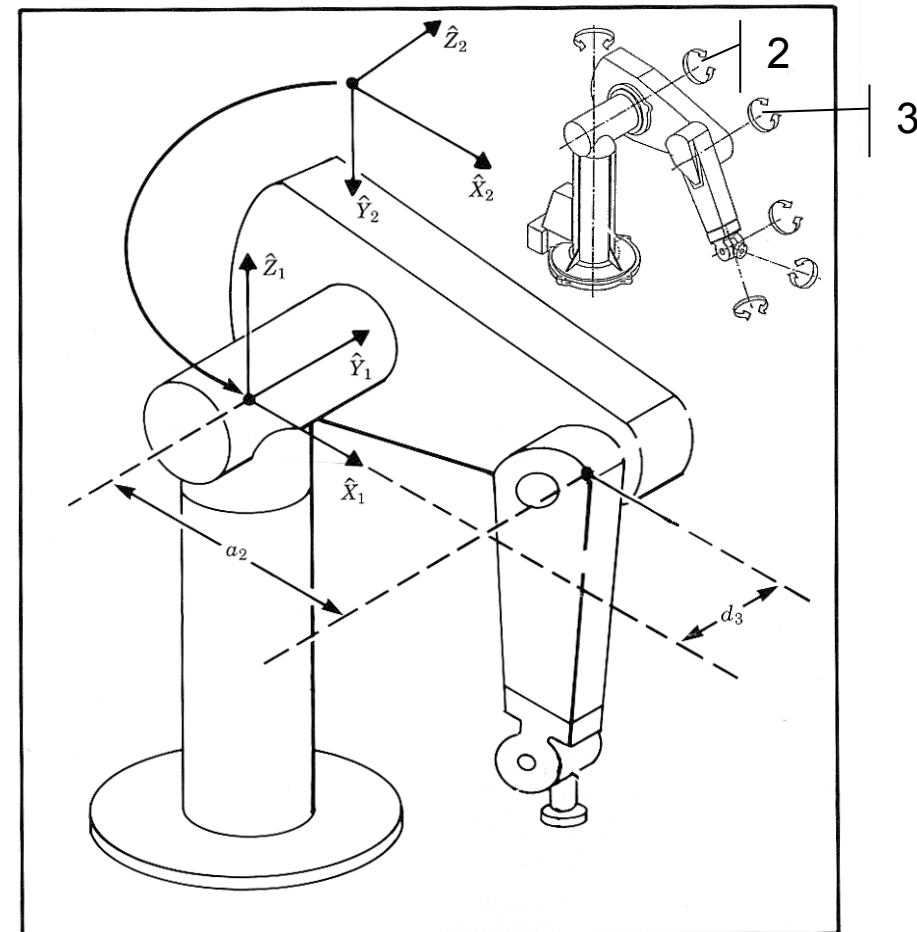
- Assign {0} to match {1} when the first joint variable is zero. Frame {0} is coincident with Frame {1}
- Assign the $\hat{Z}_1$ axis pointing along the **1st** joint axis.
- Assign the $\hat{X}_1$ axis pointing along the common perpendicular, or **if the axes intersect, assign $\hat{X}_1$ to be normal to the plane containing the two axes**
- Assign the $\hat{Y}_1$ axis to complete a right hand coordinate system.



Forward Manipulator Kinematics

Example: Frame Assignments - Frame {2}

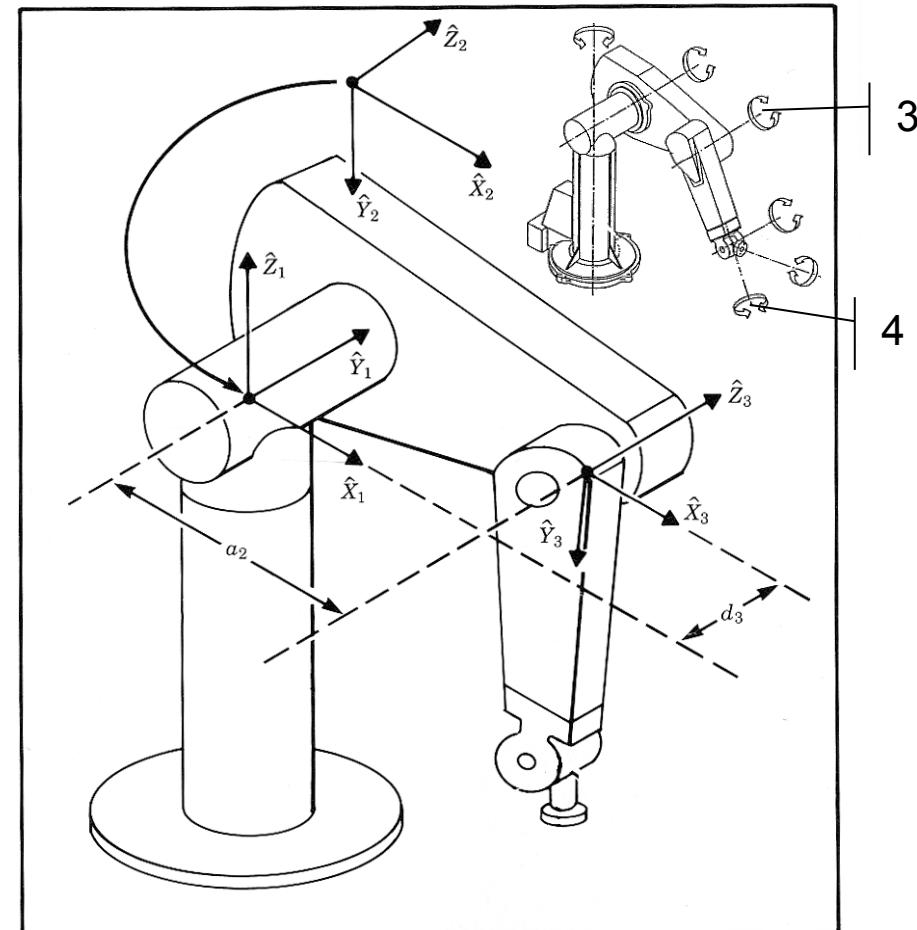
- Assign the $\hat{Z}_2$ axis pointing along the **2nd** joint axis.
- Assign the $\hat{X}_2$ axis pointing along the common perpendicular
- Assign the $\hat{Y}_2$ axis to the complete a right hand coordinate system.



Forward Manipulator Kinematics

Example: Frame Assignments - Frame {3}

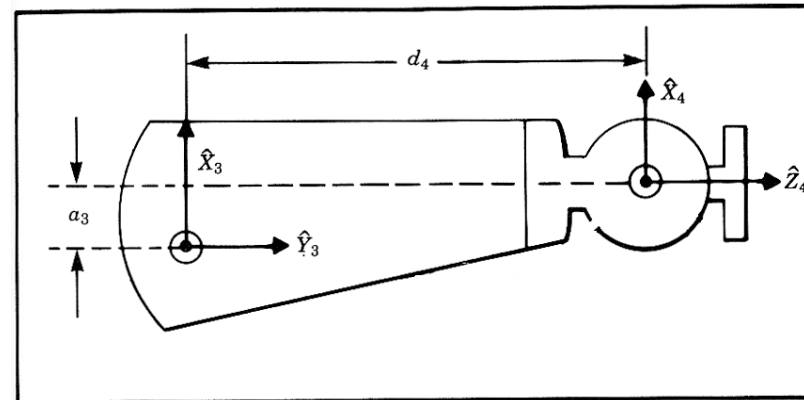
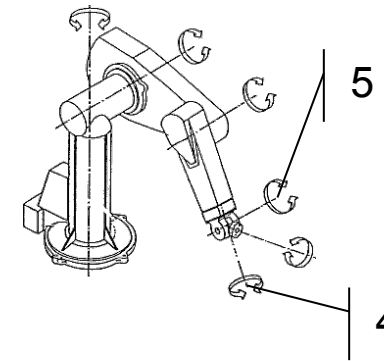
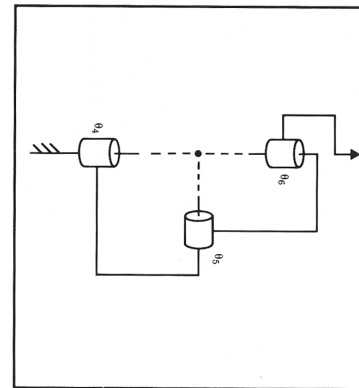
- Assign the $\hat{Z}_3$ axis pointing along the **3rd** joint axis.
- Assign the $\hat{X}_3$ axis pointing along the common perpendicular
- Assign the $\hat{Y}_3$ axis to the complete a right hand coordinate system.



Forward Manipulator Kinematics

Example: Frame Assignments - Frame {4}

- Assign the $\hat{Z}_4$ axis pointing along the **4th** joint axis.
- Assign the $\hat{X}_4$ axis pointing along the common perpendicular **if the axes intersect, assign $\hat{X}_4$ to be normal to the plane containing the two axes**
- Assign the $\hat{Y}_4$ axis to the complete a right hand coordinate system.



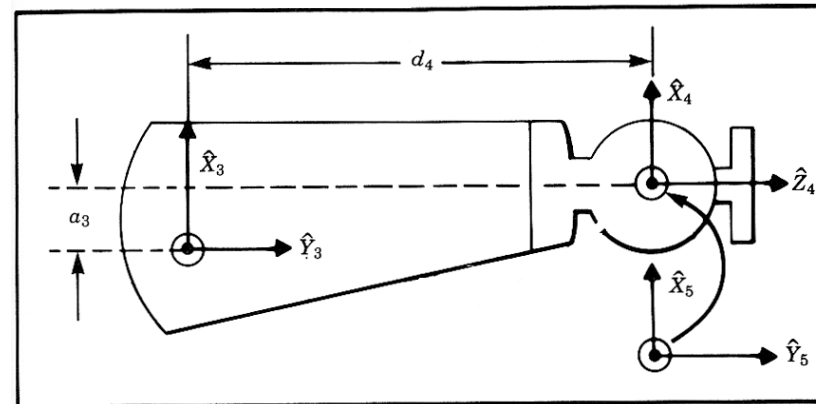
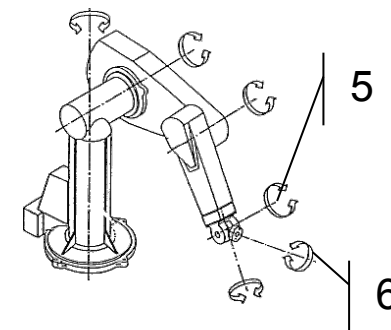
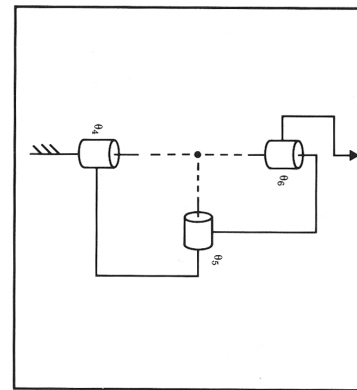
Forward Manipulator Kinematics

Example: Frame Assignments - Frame {4}, {5} ,{6}

Forward Manipulator Kinematics

Example: Frame Assignments - Frame {5}

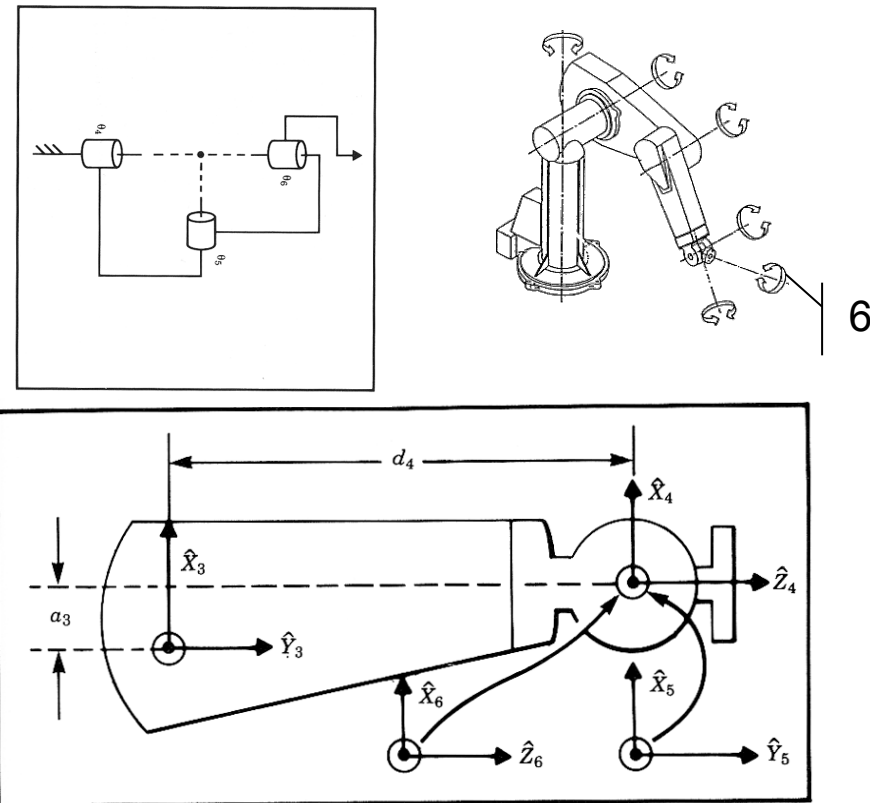
- Assign the $\hat{Z}_5$ axis pointing along the **5th** joint axis.
- Assign the $\hat{X}_5$ axis pointing along the common perpendicular **if the axes intersect, assign $\hat{X}_5$ to be normal to the plane containing the two axes**
- Assign the $\hat{Y}_5$ axis to the complete a right hand coordinate system.



Forward Manipulator Kinematics

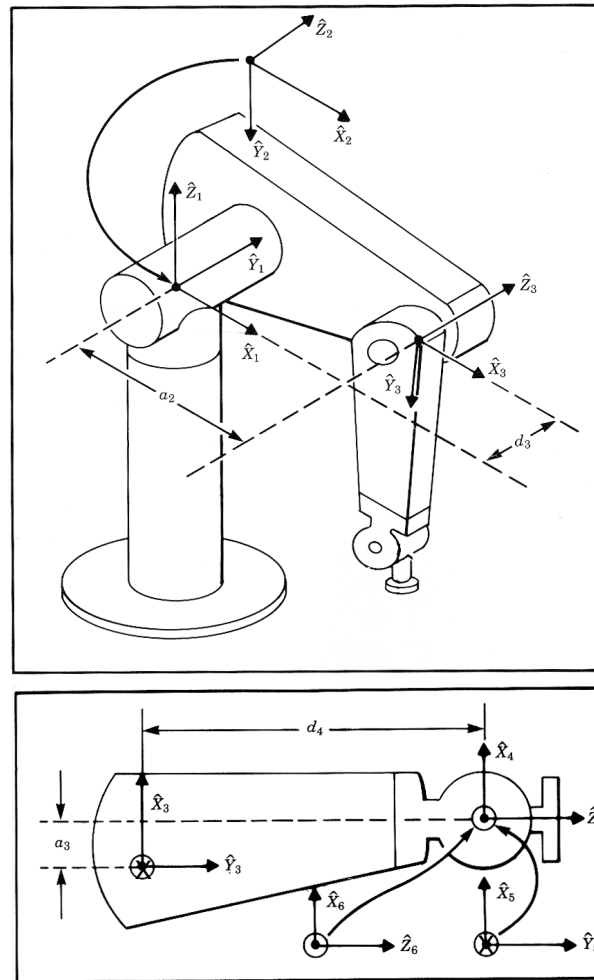
Example: Frame Assignments - Frame {6}

- Assign the $\hat{Z}_6$ axis pointing along the **6th** joint axis.
- For frame {N} ({6}) choose an origin location and $\hat{X}_6$ direction freely, but generally so as to cause as many linkage parameters as possible to be zero
- Assign the $\hat{Y}_6$ axis to the complete a right hand coordinate system.



Forward Manipulator Kinematics

Example: PUMA 560 DH Parameters



α_{i-1} - The angle between $\hat{Z}_{i-1}$ to $\hat{Z}_i$
measured about $\hat{X}_{i-1}$

$i-1$	i	α_{i-1}
0	1	0
1	2	-90
2	3	0
3	4	-90
4	5	90
5	6	-90

$$\alpha_0 = \angle \hat{Z}_0 \hat{Z}_1 \text{ about } \hat{X}_0$$

$$\alpha_1 = \angle \hat{Z}_1 \hat{Z}_2 \text{ about } \hat{X}_1$$

$$\alpha_2 = \angle \hat{Z}_2 \hat{Z}_3 \text{ about } \hat{X}_2$$

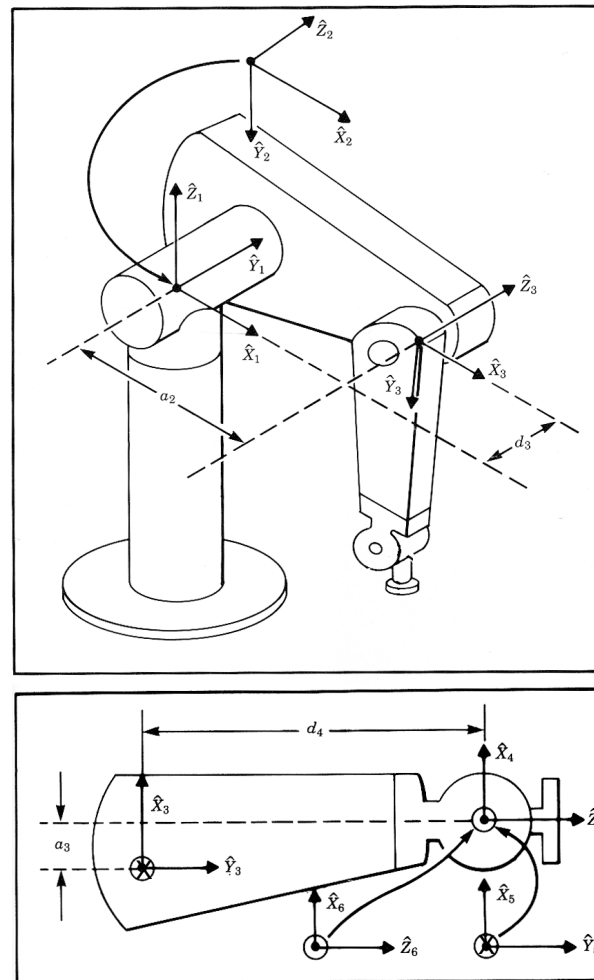
$$\alpha_3 = \angle \hat{Z}_3 \hat{Z}_4 \text{ about } \hat{X}_3$$

$$\alpha_4 = \angle \hat{Z}_4 \hat{Z}_5 \text{ about } \hat{X}_4$$

$$\alpha_5 = \angle \hat{Z}_5 \hat{Z}_6 \text{ about } \hat{X}_5$$

Forward Manipulator Kinematics

Example: PUMA 560 DH Parameters



a_{i-1} - The distance from $\hat{Z}_{i-1}$ to $\hat{Z}_i$
measured along $\hat{X}_{i-1}$

$i-1$	i	a_{i-1}
0	1	0
1	2	0
2	3	a_2
3	4	a_3
4	5	0
5	6	0

$a_0 = \text{Dist. } \hat{Z}_0 \hat{Z}_1 \text{ along } \hat{X}_0$

$a_1 = \text{Dist. } \hat{Z}_1 \hat{Z}_2 \text{ along } \hat{X}_1$

$a_2 = \text{Dist. } \hat{Z}_2 \hat{Z}_3 \text{ along } \hat{X}_2$

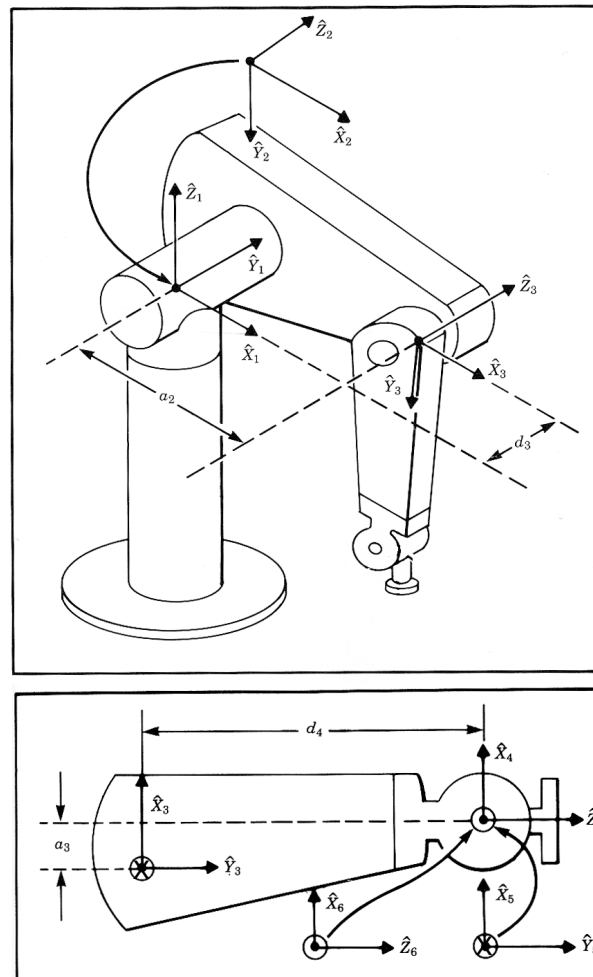
$a_3 = \text{Dist. } \hat{Z}_3 \hat{Z}_4 \text{ along } \hat{X}_3$

$a_4 = \text{Dist. } \hat{Z}_4 \hat{Z}_5 \text{ along } \hat{X}_4$

$a_5 = \text{Dist. } \hat{Z}_5 \hat{Z}_6 \text{ along } \hat{X}_5$

Forward Manipulator Kinematics

Example: PUMA 560 DH Parameters

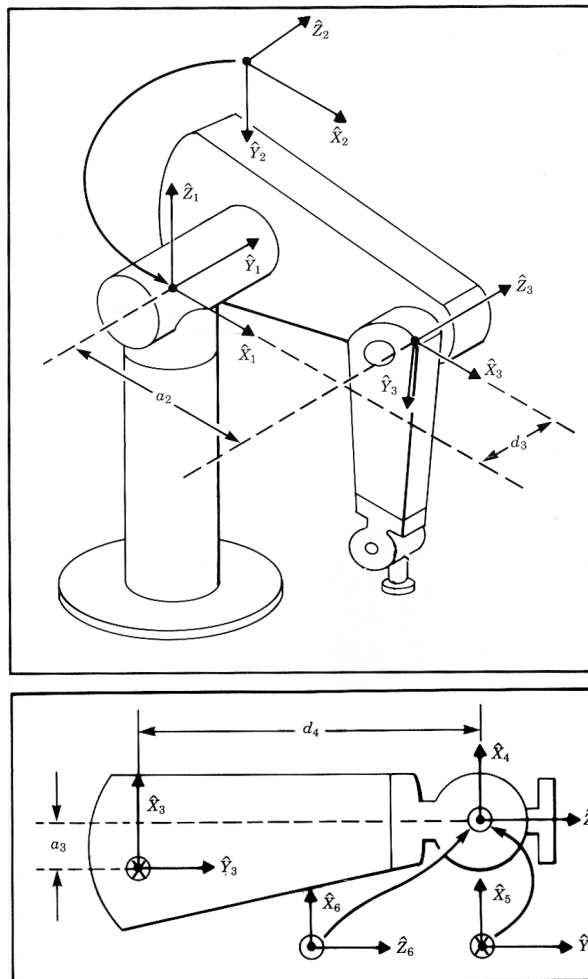


d_i - The distance from $\hat{X}_{i-1}$ to $\hat{X}_i$
measured along $\hat{Z}_i$

$i-1$	i	d_i	
0	1	0	$d_1 = \text{Dist. } \hat{X}_0 \hat{X}_1 \text{ along } Z_1$
1	2	0	$d_2 = \text{Dist. } \hat{X}_1 \hat{X}_2 \text{ along } Z_2$
2	3	d_3	$d_3 = \text{Dist. } \hat{X}_2 \hat{X}_3 \text{ along } Z_3$
3	4	d_4	$d_4 = \text{Dist. } \hat{X}_3 \hat{X}_4 \text{ along } Z_4$
4	5	0	$d_5 = \text{Dist. } \hat{X}_4 \hat{X}_5 \text{ along } Z_5$
5	6	0	$d_6 = \text{Dist. } \hat{X}_5 \hat{X}_6 \text{ along } Z_6$

Forward Manipulator Kinematics

Example: PUMA 560 DH Parameters



θ_i - The angle between $\hat{X}_{i-1}$ to $\hat{X}_i$
measured about $\hat{Z}_i$

$i-1$	i	θ_i
0	1	θ_1
1	2	θ_2
2	3	θ_3
3	4	θ_4
4	5	θ_5
5	6	θ_6

$$\theta_1 = \angle \hat{X}_0 \hat{X}_1 \text{ about } \hat{Z}_1$$

$$\theta_2 = \angle \hat{X}_1 \hat{X}_2 \text{ about } \hat{Z}_2$$

$$\theta_3 = \angle \hat{X}_2 \hat{X}_3 \text{ about } \hat{Z}_3$$

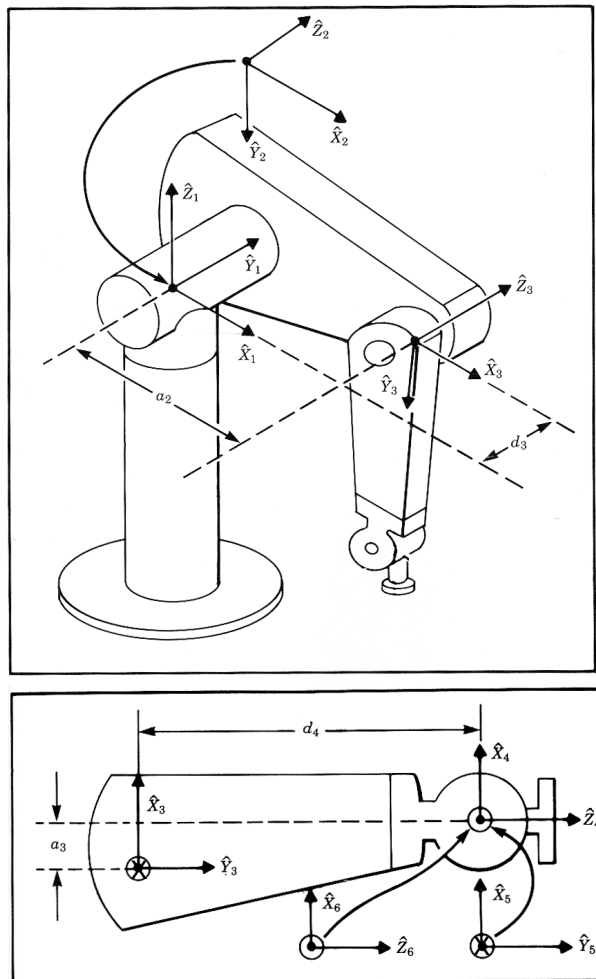
$$\theta_4 = \angle \hat{X}_3 \hat{X}_4 \text{ about } \hat{Z}_4$$

$$\theta_5 = \angle \hat{X}_4 \hat{X}_5 \text{ about } \hat{Z}_5$$

$$\theta_6 = \angle \hat{X}_5 \hat{X}_6 \text{ about } \hat{Z}_6$$

Forward Manipulator Kinematics

Example: PUMA 560 DH Parameters



i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90°	0	0	θ_2
3	0	a_2	d_3	θ_3
4	-90°	a_3	d_4	θ_4
5	90°	0	0	θ_5
6	-90°	0	0	θ_6

Forward Manipulator Kinematics

Example: PUMA 560 Link Transformations

$${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90°	0	0	θ_2
3	0	a_2	d_3	θ_3
4	-90°	a_3	d_4	θ_4
5	90°	0	0	θ_5
6	-90°	0	0	θ_6

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_2 & -c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4T = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & a_3 \\ 0 & 0 & 1 & d_4 \\ -s\theta_4 & -c\theta_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^4_5T = \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_5 & c\theta_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^5_6T = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_6 & -c\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Forward Manipulator Kinematics

Example: PUMA 560 Kinematics Equations

- The kinematics equations of PUMA 560 specify how to compute the position & orientation of frame {6} (tool) relative to frame {0} (base) of the robot. These are the basic equations for all kinematic analysis of this manipulator.

$${}^0T_6 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5 {}^5T_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Notations

$$\cos(\theta_1) = c\theta_1 = c_1$$

$$\cos(\theta_1 + \theta_2) = c_{12} = c_1 c_2 - s_1 s_2$$

$$\sin(\theta_1 + \theta_2) = s_{12} = c_1 s_2 + s_1 c_2$$

$$r_{11} = c_1 [c_{23}(c_4 c_5 c_6 - s_4 s_6) - s_{23} s_5 c_6] + s_1 (s_4 c_5 c_6 + c_4 s_6),$$

$$r_{21} = s_1 [c_{23}(c_4 c_5 c_6 - s_4 s_6) - s_{23} s_5 c_6] - c_1 (s_4 c_5 c_6 + c_4 s_6),$$

$$r_{31} = -s_{23}(c_4 c_5 c_6 - s_4 s_6) - c_{23} s_5 c_6,$$

$$r_{12} = c_1 [c_{23}(-c_4 c_5 s_6 - s_4 c_6) + s_{23} s_5 s_6] + s_1 (c_4 c_6 - s_4 c_5 s_6),$$

$$r_{22} = s_1 [c_{23}(-c_4 c_5 s_6 - s_4 c_6) + s_{23} s_5 s_6] - c_1 (c_4 c_6 - s_4 c_5 s_6),$$

$$r_{32} = -s_{23}(-c_4 c_5 s_6 - s_4 c_6) + c_{23} s_5 s_6,$$

$$r_{13} = -c_1 (c_{23} c_4 s_5 + s_{23} c_5) - s_1 s_4 s_5,$$

$$r_{23} = -s_1 (c_{23} c_4 s_5 + s_{23} c_5) + c_1 s_4 s_5,$$

$$r_{33} = s_{23} c_4 s_5 - c_{23} c_5,$$

$$p_x = c_1 [a_2 c_2 + a_3 c_{23} - d_4 s_{23}] - d_3 s_1,$$

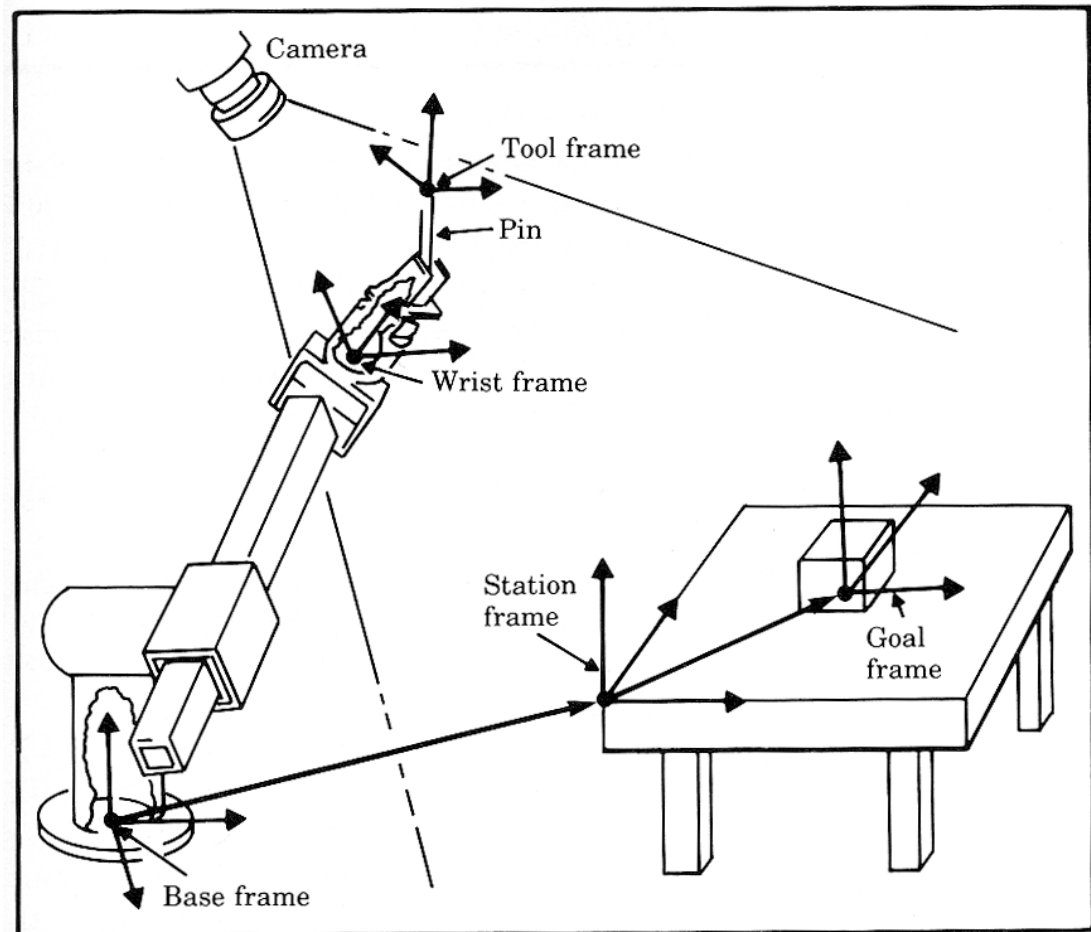
$$p_y = s_1 [a_2 c_2 + a_3 c_{23} - d_4 s_{23}] + d_3 c_1,$$

$$p_z = -a_3 s_{23} - a_2 s_2 - d_4 c_{23}.$$

Forward Manipulator Kinematics

Frames with Standard Names

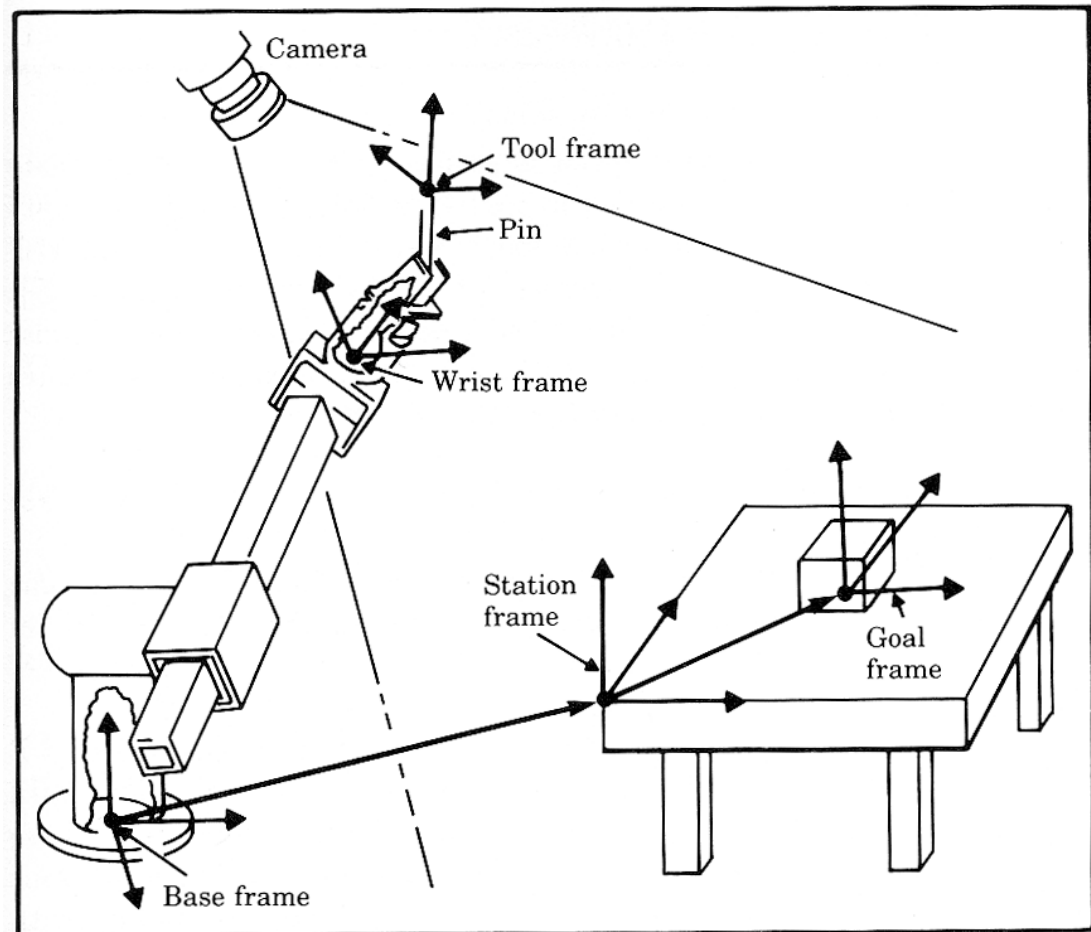
- **Base Frame {B}** - {B} is located at the base of the manipulator affixed to the nonmoving part of the robot (another name for frame {0})
- **Station Frame {S}** - {S} is located in a task relevant location (e.g. at the corner of the table upon the which the robot is to work). From the user perspective {S} is the universe frame (task frame or world frame) and all action of the robot are made relative to it. The station frame {S} is always specify with respect to the base frame {B}, i.e. ${}^B T_S$



Forward Manipulator Kinematics

Frames with Standard Names

- **Wrist Frame {W}** - {W} is affixed to the last link of the manipulator - the wrist (another name for frame {N}). The wrist frame {W} is defined relative to the base frame i.e. ${}^B T_W = {}^0 T_N$
- **Tool Frame {T}** - {T} is affixed to the end of any tool the robot happens to be holding. When the hand is empty, {T} is located with its origin between the fingertips of the robot. The tool frame {T} is always specified with respect to the wrist frame {W} i.e. ${}^W T_T$
- **Goal Frame {G}** - {G} is describing the location to which the robot is about to move the tool. At the end of the robot motion the tool frame {T} is about to coincide with the the goal frame {G}. The goal frame is always specified with respect to the station frame {S} i.e. ${}^S T_G$

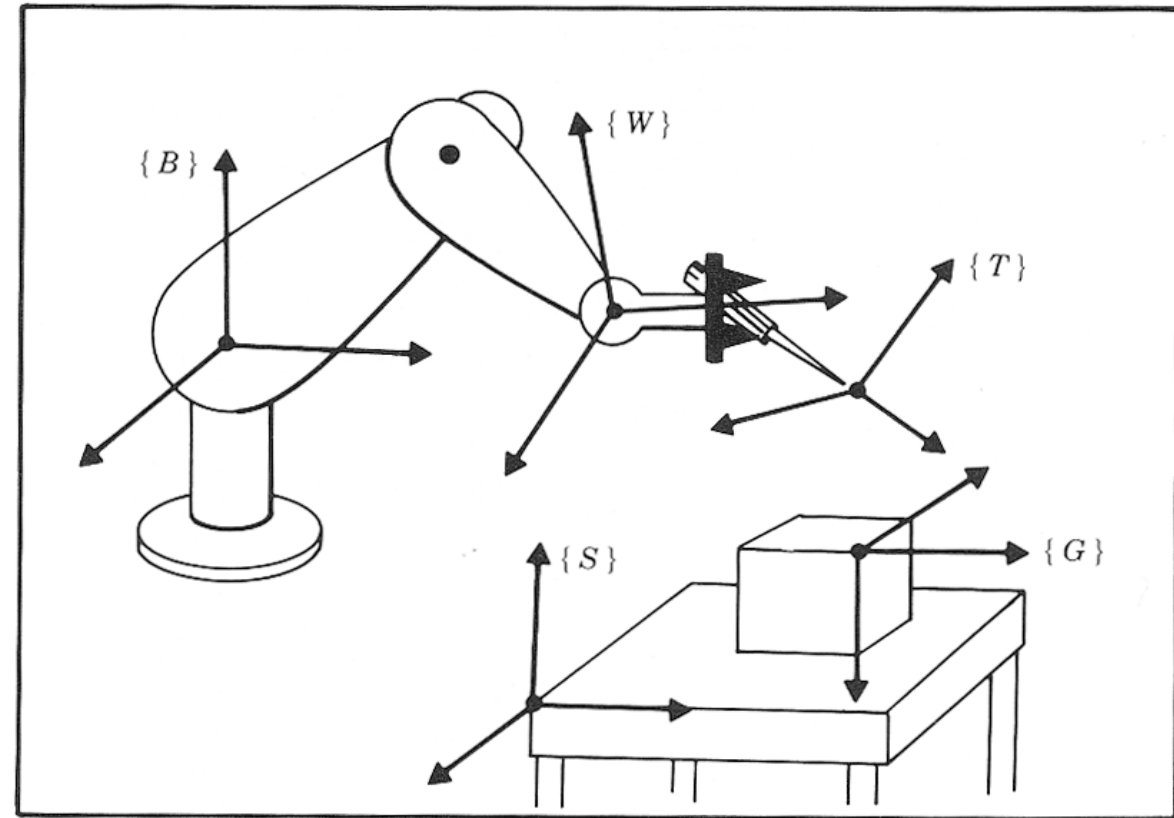


Forward Manipulator Kinematics

Where is the tool?

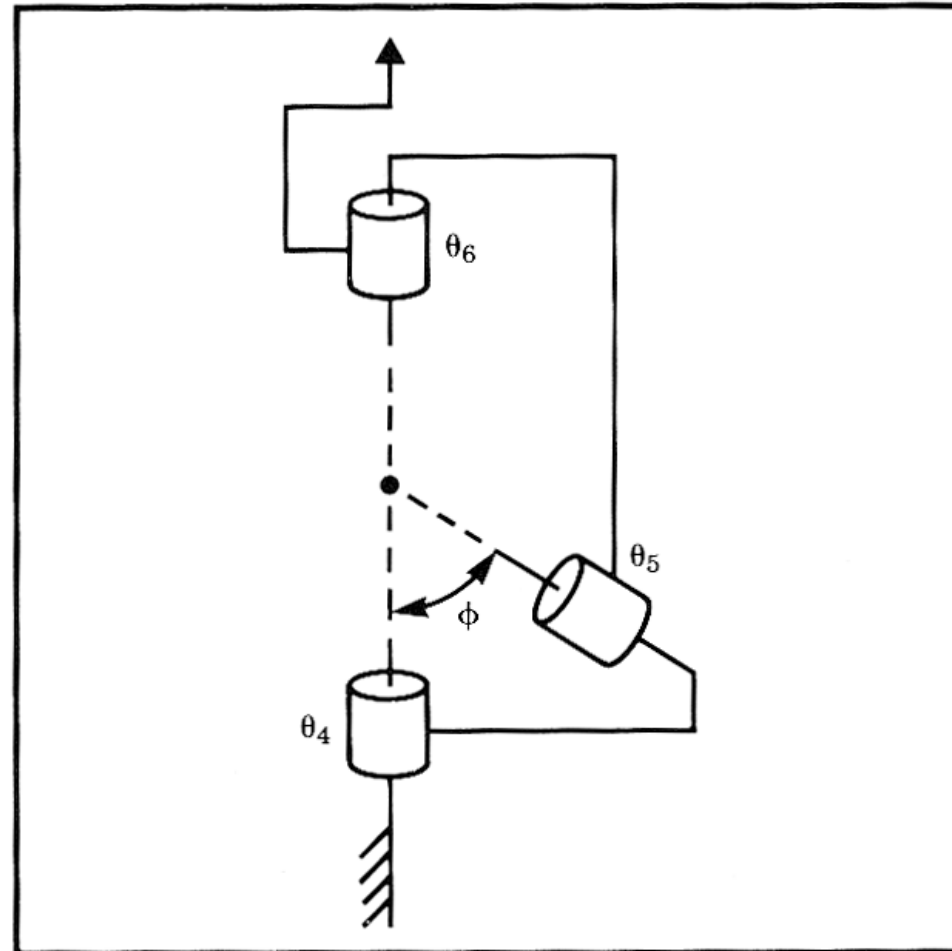
- **Problem:**
Calculate the transformation matrix of the tool frame $\{T\}$ relative to the station frame $\{S\}$ - ${}^S_T T$
- **Solution:**
Cartesian Transformation

$${}^S_T T = {}^S_B T^{-1} {}^B_W T {}^W_T T$$



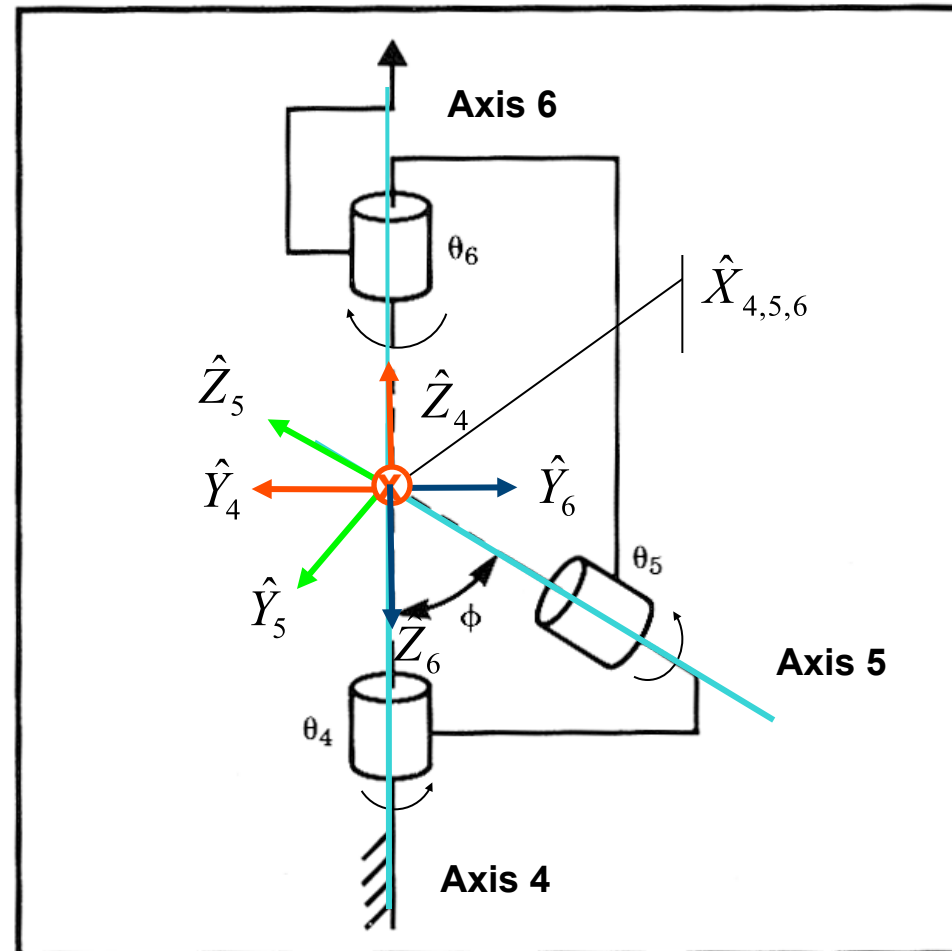
Forward Manipulator Kinematics

Example: 3R Wrist



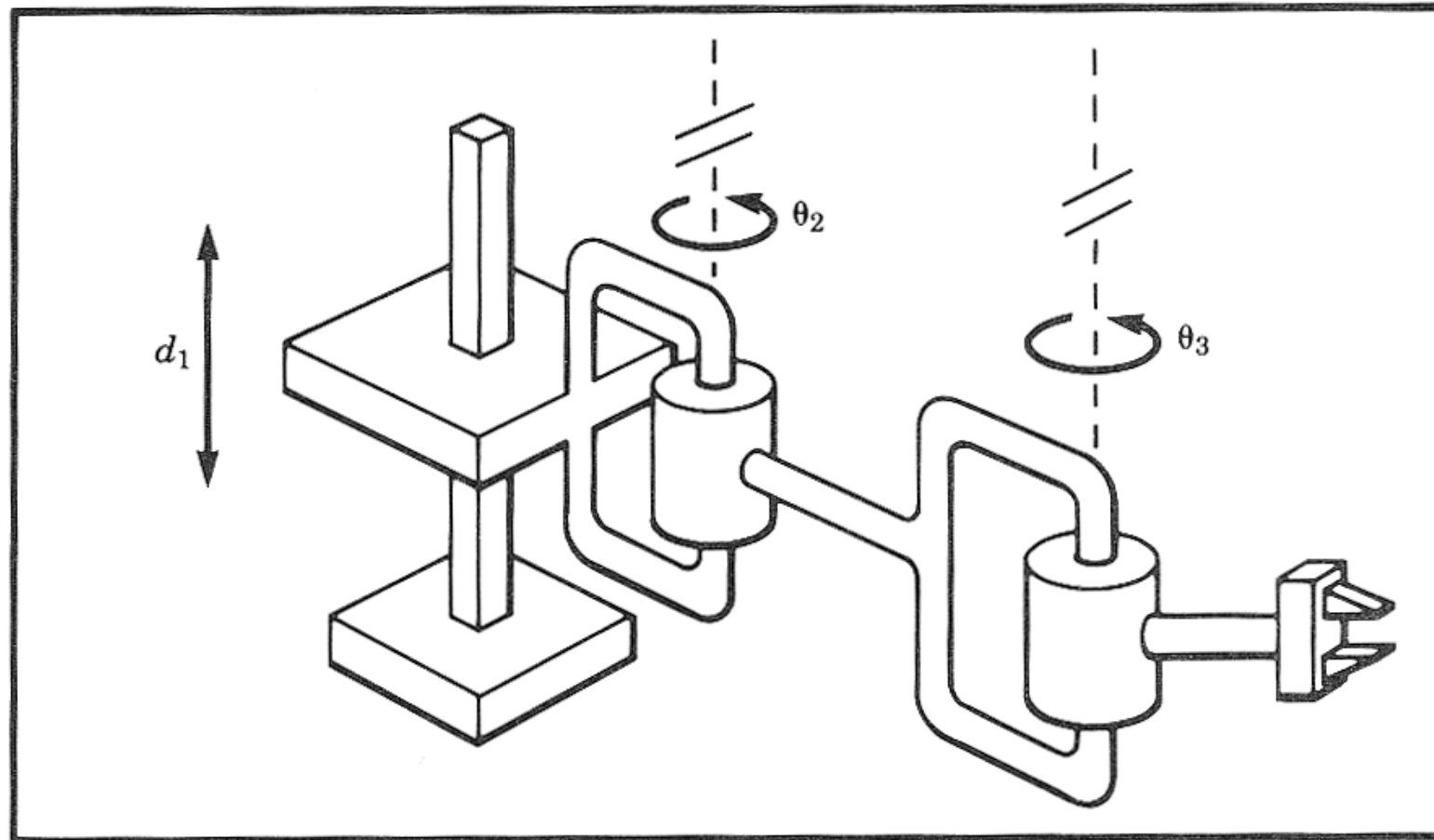
Forward Manipulator Kinematics

Example: 3R Wrist



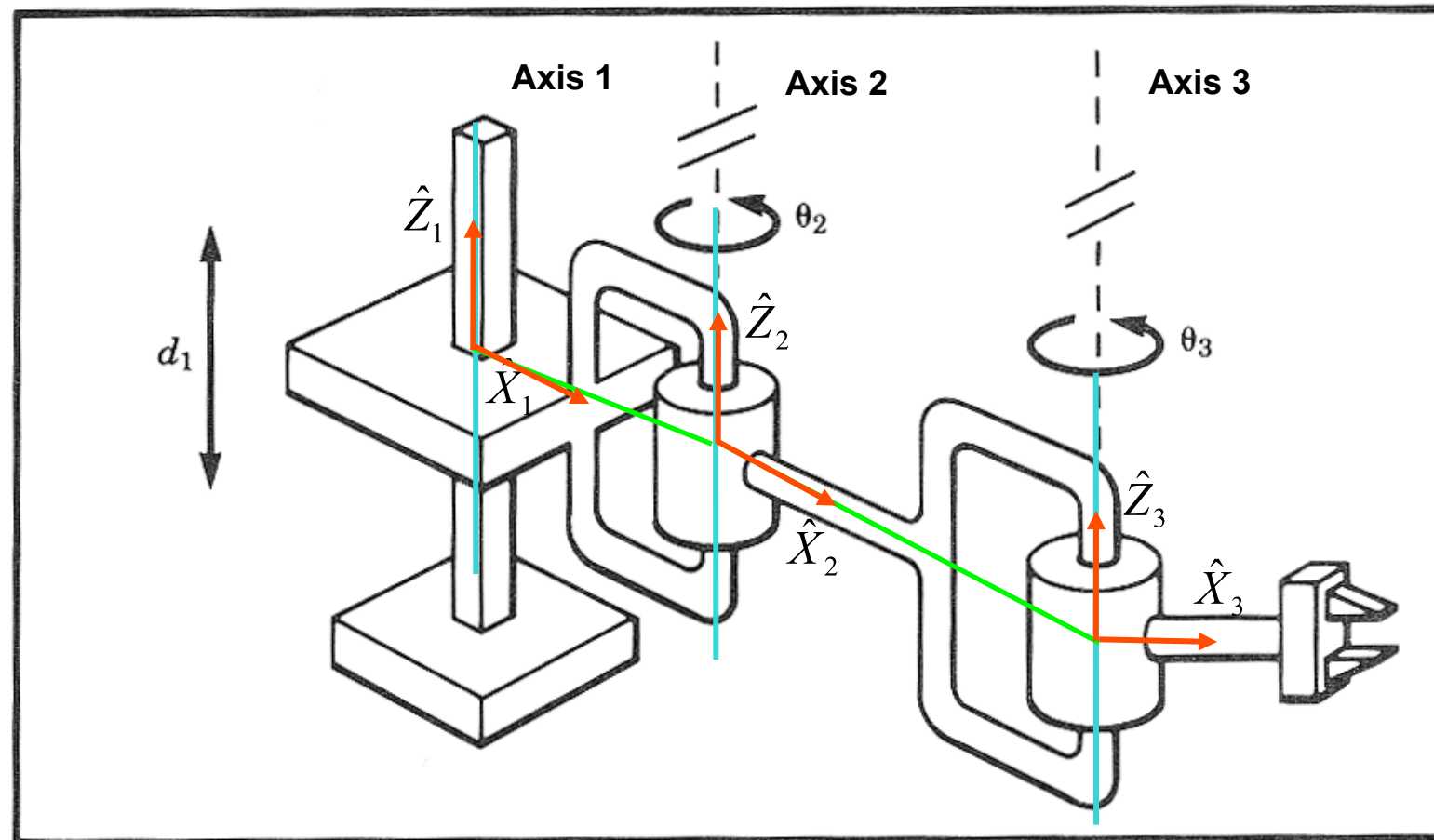
Forward Manipulator Kinematics

Example: PRR



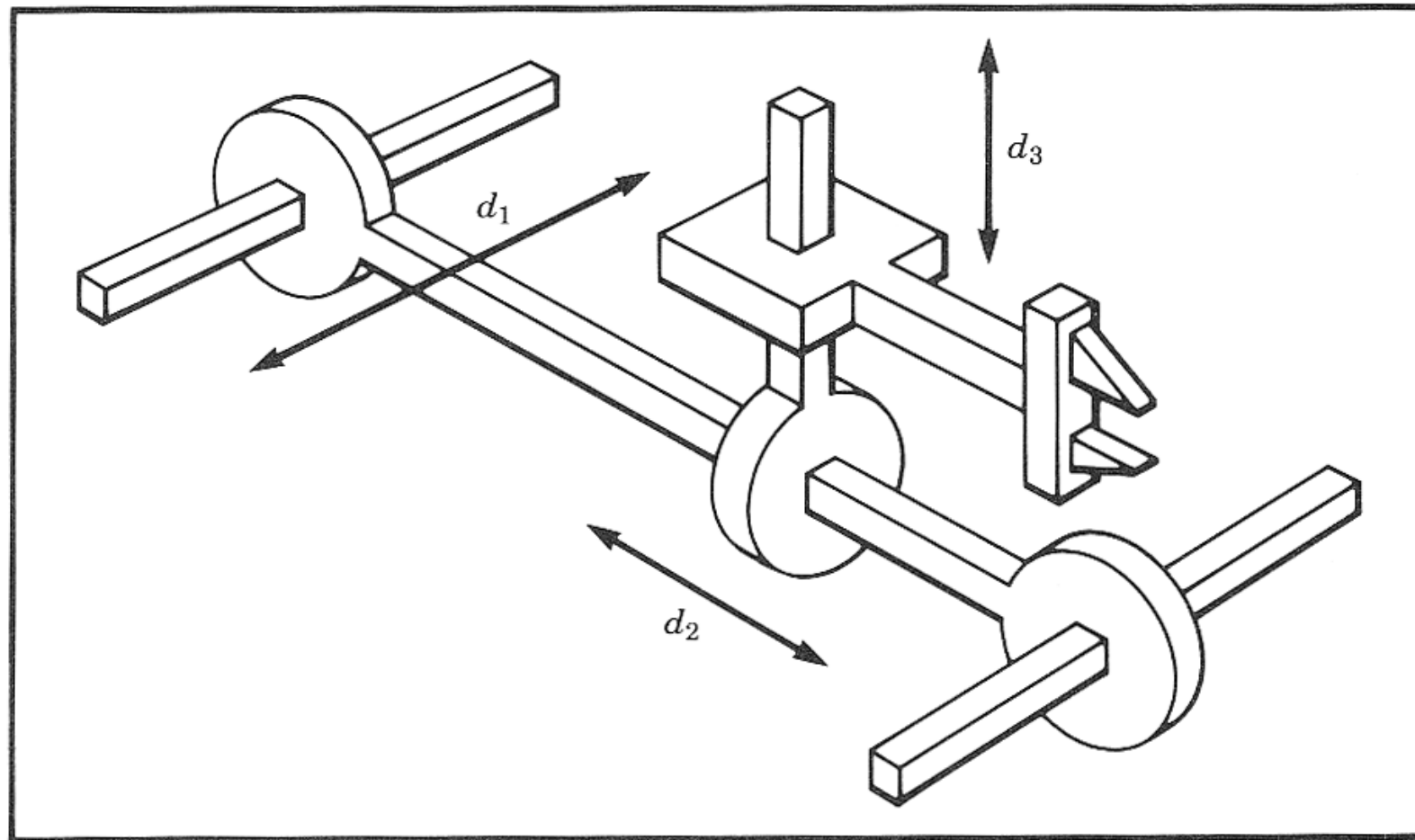
Forward Manipulator Kinematics

Example: PRR



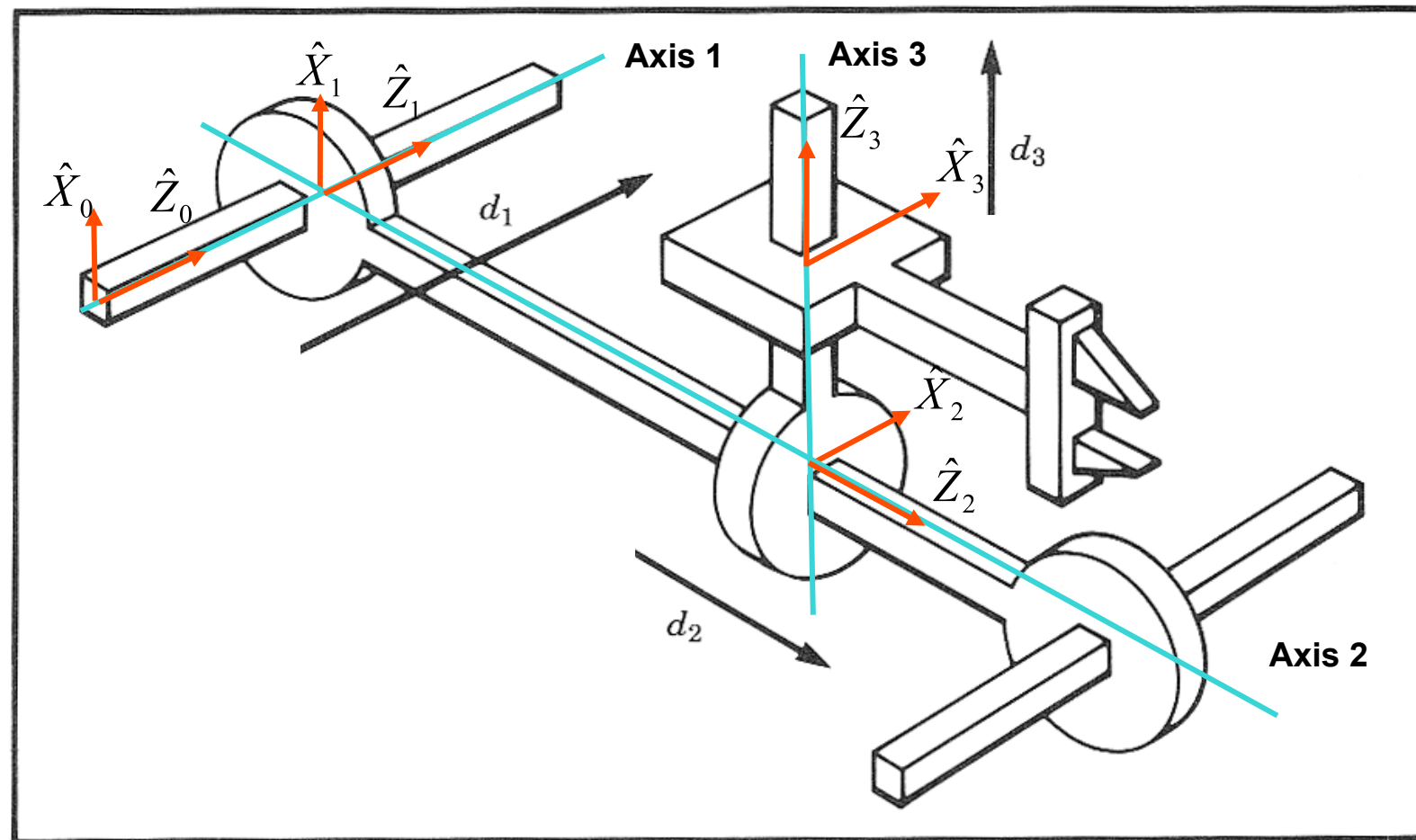
Forward Manipulator Kinematics

Example: 3P



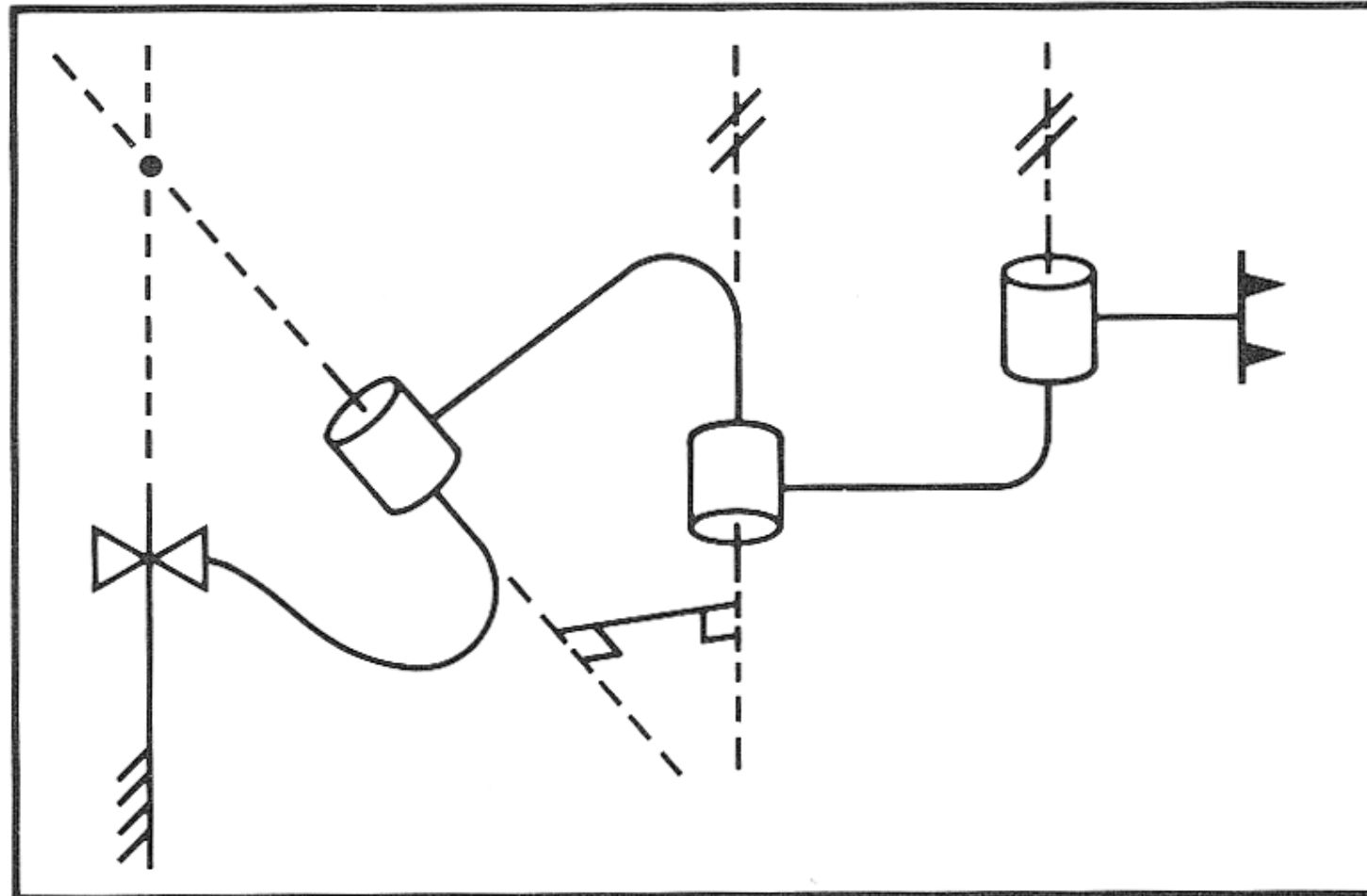
Forward Manipulator Kinematics

Example: 3P



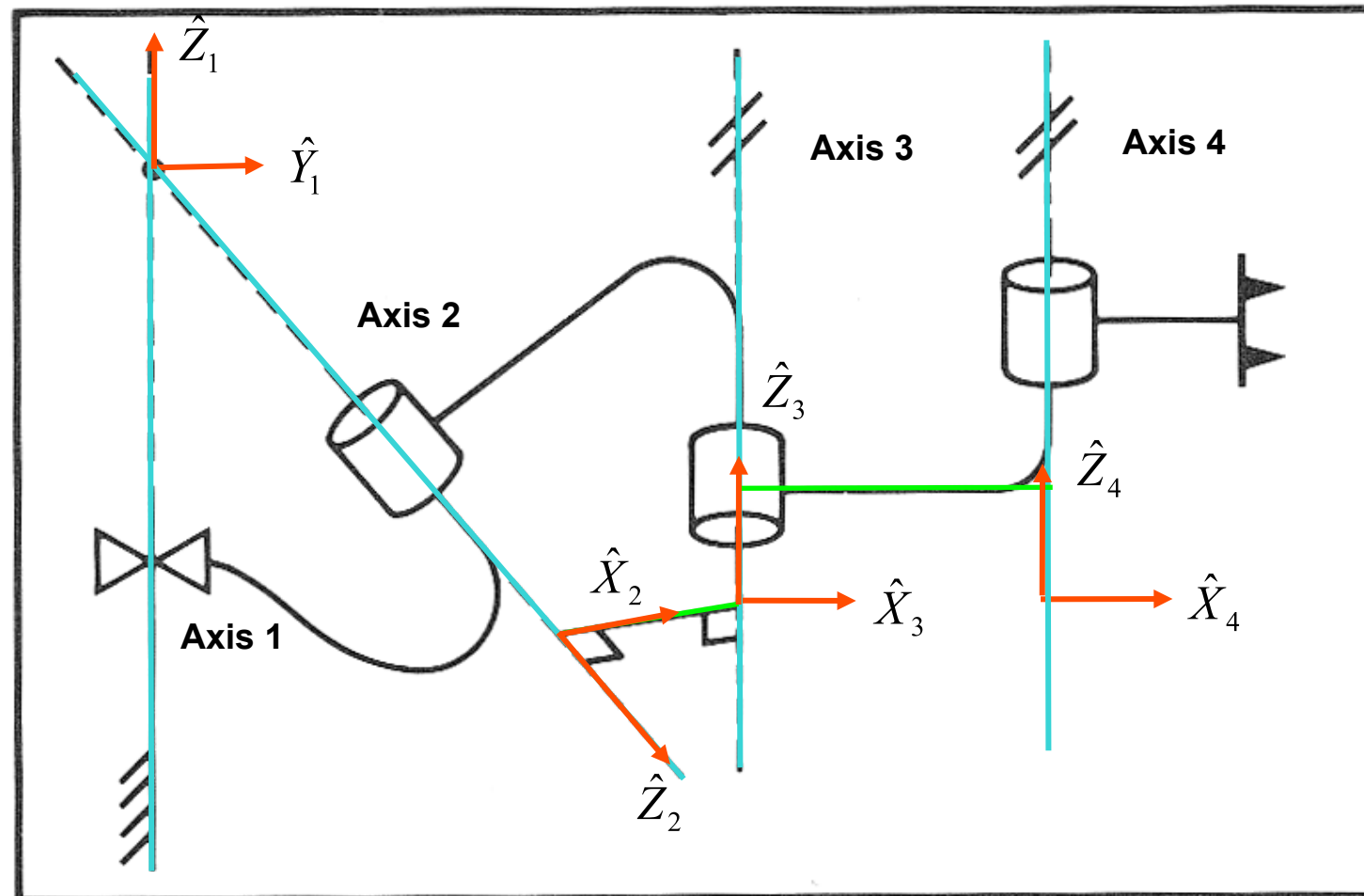
Forward Manipulator Kinematics

Example: PRRR



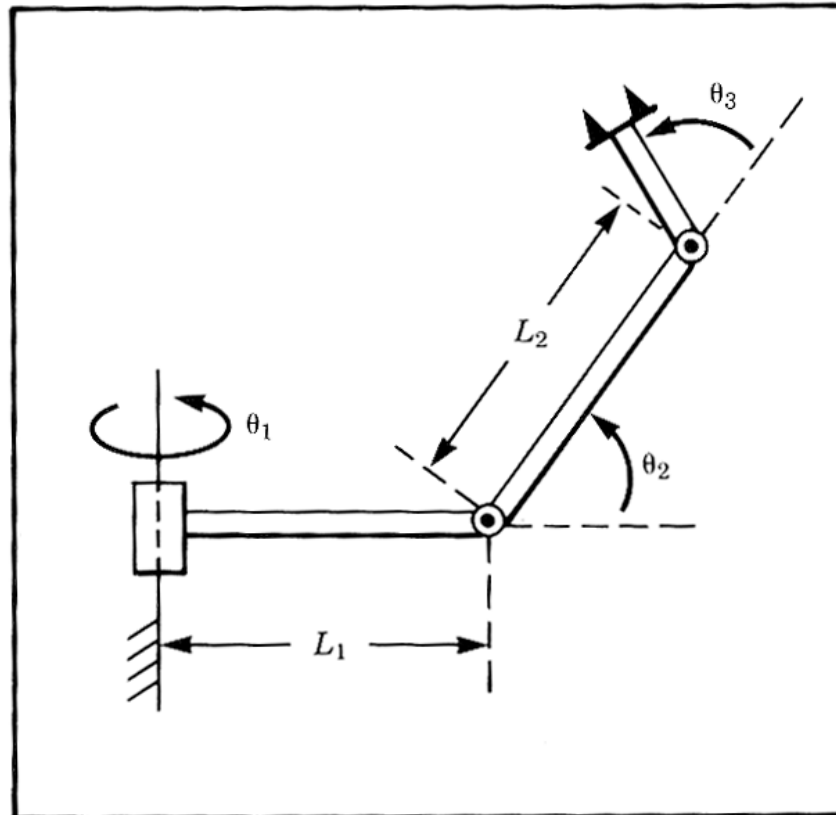
Forward Manipulator Kinematics

Example: PRRR



Forward Manipulator Kinematics

Example: 3R

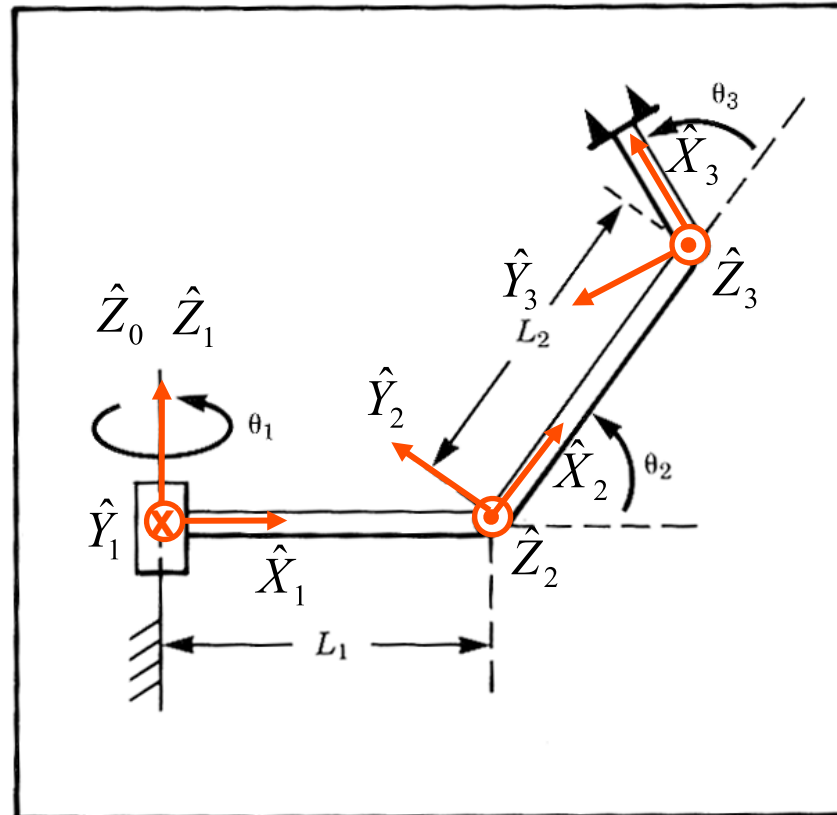


$i-1$	i	α_{i-1}	a_{i-1}	d_i	θ_i
0	1				
1	2				
2	3				

$${}^B_W T$$

Forward Manipulator Kinematics

Example: 3R

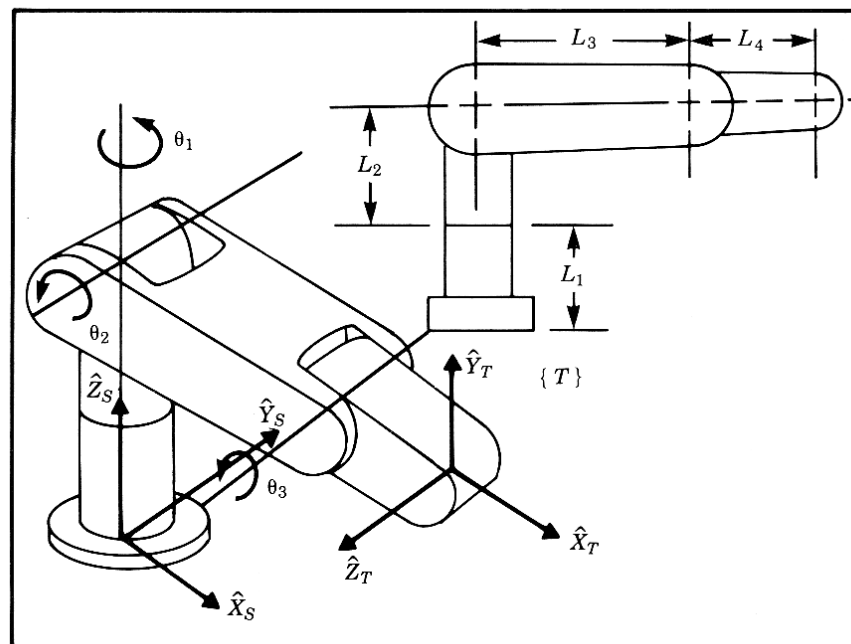


$i-1$	i	α_{i-1}	a_{i-1}	d_i	θ_i
0	1	0	0	0	θ_1
1	2	90	L_1	0	θ_2
2	3	0	L_2	0	θ_3

$${}^B_T = {}^0_3T = {}^0_1T {}^1_2T {}^2_3T$$

Forward Manipulator Kinematics

Example: 3R

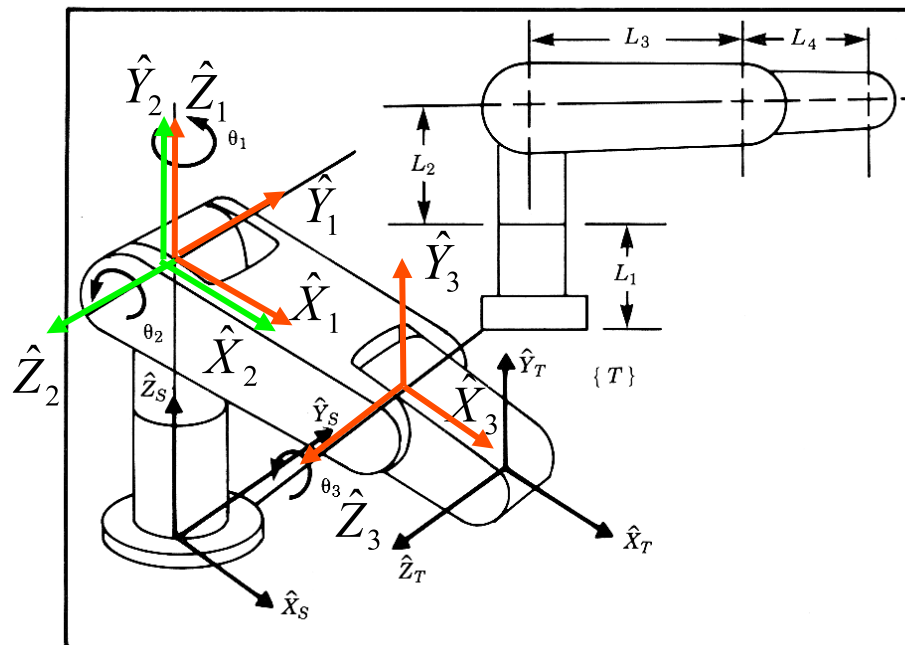


$i-1$	i	α_{i-1}	a_{i-1}	d_i	θ_i
0	1				
1	2				
2	3				
3	4				

$${}^B_T T$$

Forward Manipulator Kinematics

Example: 3R

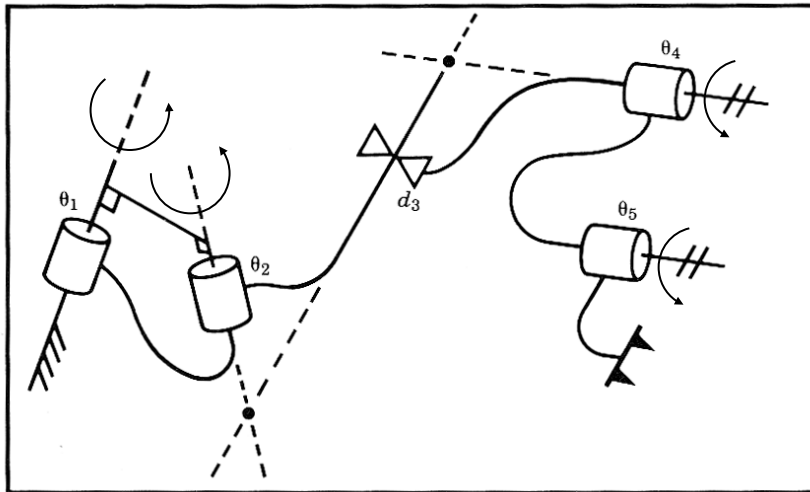


$i-1$	i	α_{i-1}	a_{i-1}	d_i	θ_i
0	1	0	0	$L_1 + L_2$	θ_1
1	2	90	0	0	θ_2
2	3	0	L_3	0	θ_3
3	4	0	L_4	0	0

$${}^S_T T = {}^0_4 T = {}^0_1 T {}^1_2 T {}^2_3 T {}^3_4 T$$

Forward Manipulator Kinematics

Example: RRP2R (2RP2R)

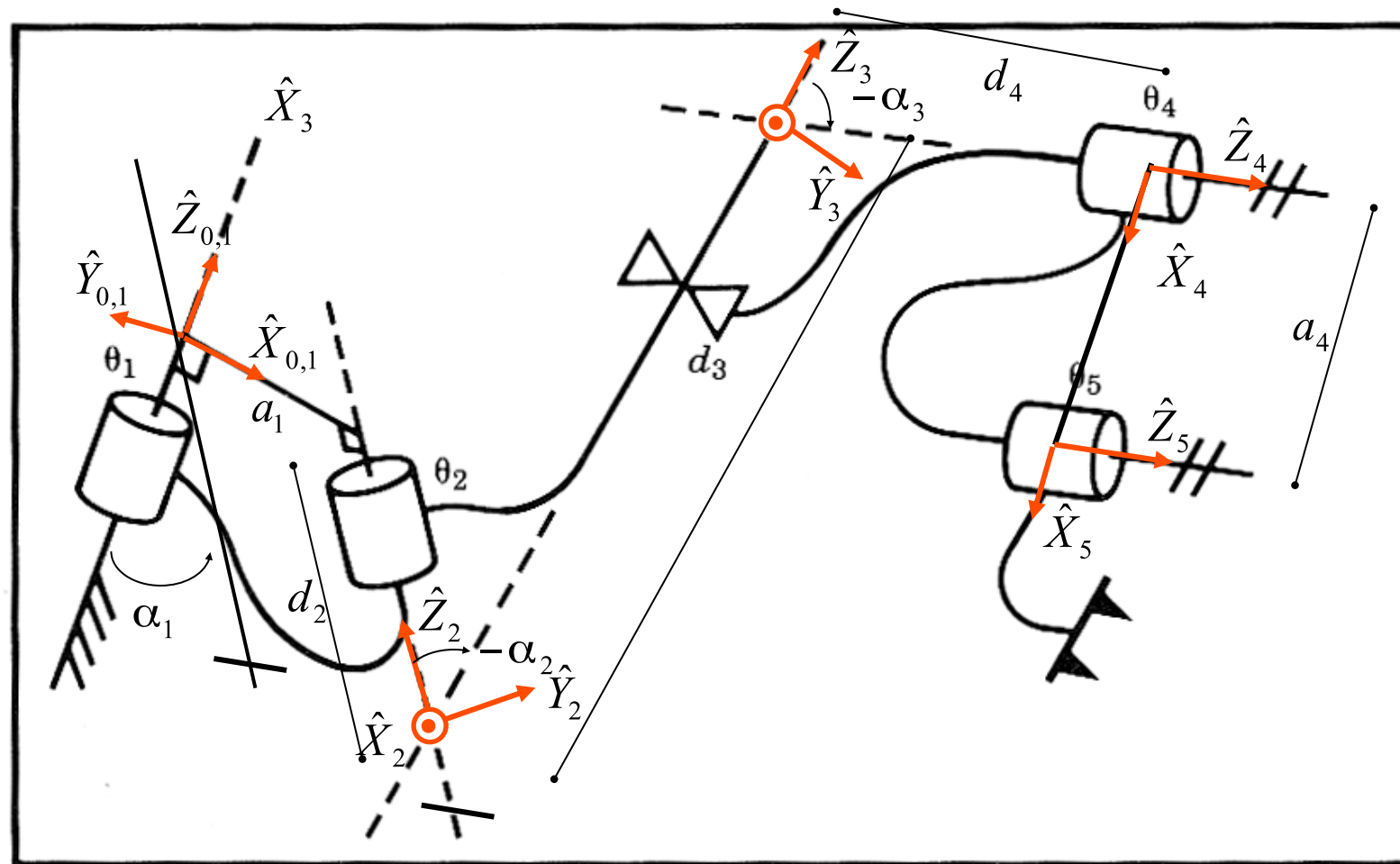


$i-1$	i	α_{i-1}	a_{i-1}	d_i	θ_i
0	1				
1	2				
2	3				
3	4				
4	5				

$${}^B_w T$$

Forward Manipulator Kinematics

Example: 2RP2R



Forward Manipulator Kinematics

Example: 2RP2R

$i-1$	i	α_{i-1}	a_{i-1}	d_i	θ_i
0	1	0	0	0	θ_1
1	2	α_1	a_1	$-d_2$	θ_2
2	3	$-\alpha_2$	0	d_3	0
3	4	$-\alpha_3$	0	d_4	θ_4
4	5	0	a_4	0	θ_5

$${}^B_W T = {}^0_5 T = {}^0_1 T {}^1_2 T {}^2_3 T {}^3_4 T {}^4_5 T$$

Forward Manipulator Kinematics

Summary: Link Frame Attachment Procedure

Joint Axes

Identify the joint axes and imagine (or draw) infinite lines along them. For the following steps below, consider two of these neighboring lines (at axes i and $i+1$)

Frame Origin

Identify the common perpendicular between them, or point of intersection. At the point of intersection, or at the point where the common perpendicular meets the i th axis, assign the link frame origin.

$\hat{Z}_i$ Axis

Assign the $\hat{Z}_i$ axis pointing along the i th joint axis.

$\hat{X}_i$ Axis

Assign the $\hat{X}_i$ axis pointing along the common perpendicular, or if the axes $\hat{Z}_{i+1}$ and $\hat{Z}_i$ intersect, assign $\hat{X}_i$ to be normal to the plane containing the two $\hat{Z}_i$ axes.

$\hat{Y}_i$ Axis

Assign the $\hat{Y}_i$ axis to complete a right hand coordinate system.

Frame {0} and Frame {1}

Assign {0} to match {1} when the first joint variable is zero. For {N} choose an origin location and $\hat{X}_N$ direction freely, but generally so as to cause as many linkage parameters as possible to be zero

Forward Manipulator Kinematics

Summary: DH Parameters

Link Length a_{i-1}

The *distance* from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ measured along $\hat{X}_{i-1}$

Link Twist α_{i-1}

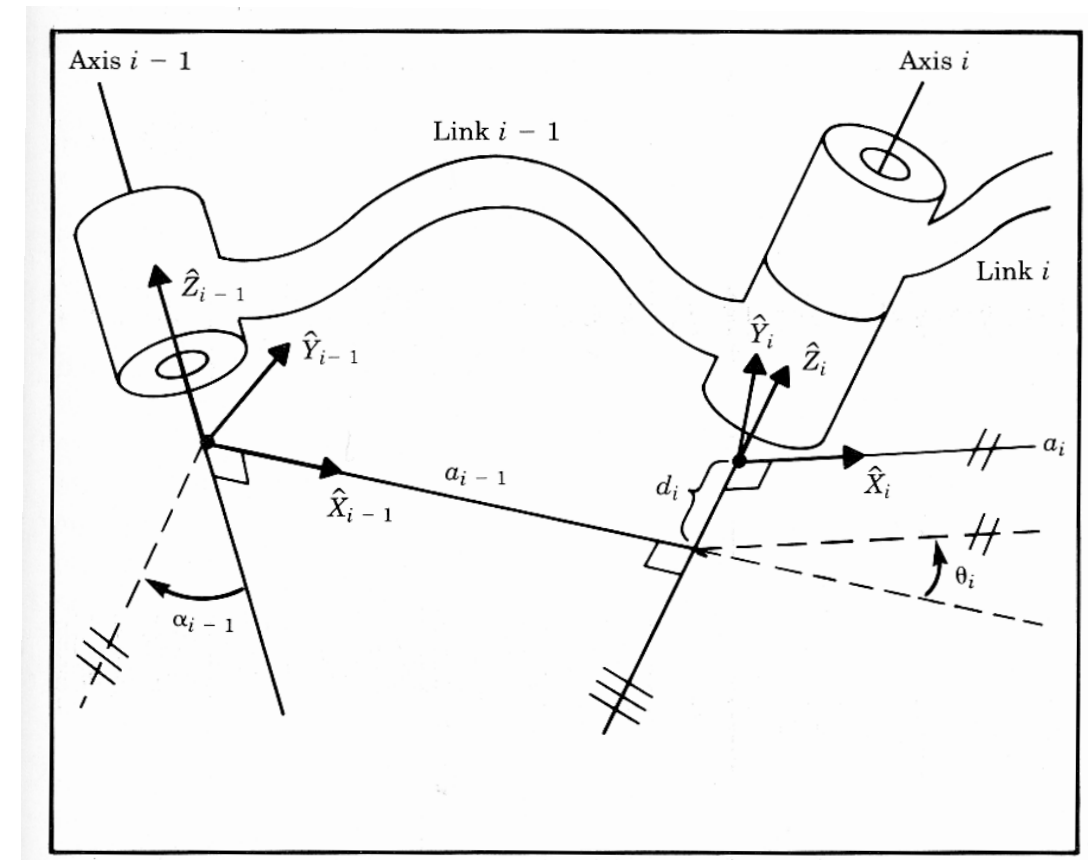
The *angle* between $\hat{Z}_{i-1}$ and $\hat{Z}_i$ measured about $\hat{X}_{i-1}$

Link Offset d_i

The *distance* from $\hat{X}_{i-1}$ to $\hat{X}_i$ measured along $\hat{Z}_i$

Link Angle θ_i

The *angle* between $\hat{X}_{i-1}$ and $\hat{X}_i$ measured about $\hat{Z}_i$

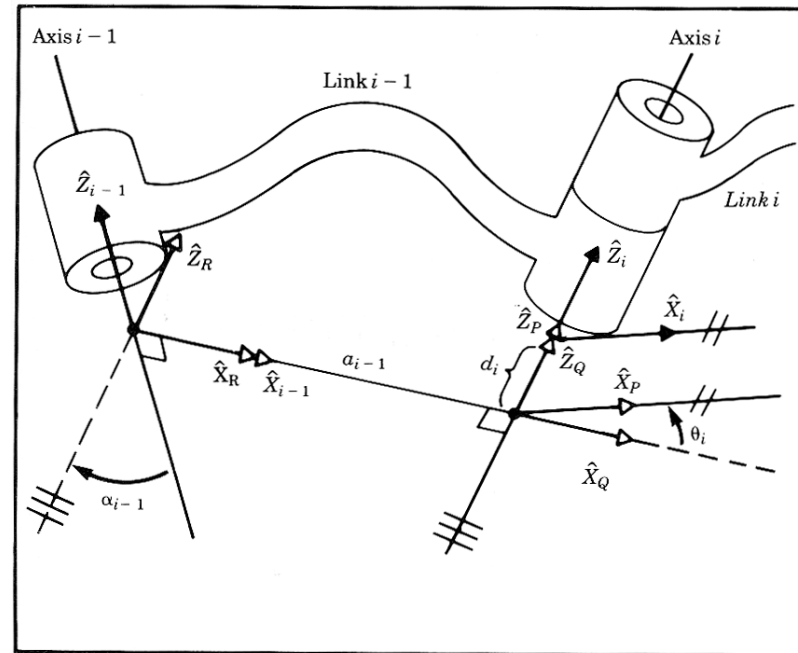


Note: $a_i \geq 0$

α_i d_i θ_i are signed quantities

Forward Manipulator Kinematics

Summary: Link Transformations



$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_N T = {}^0_1 T {}^1_2 T {}^2_3 T \dots {}^{N-1}_N T$$